Team MUTANS University of Padua

Project C.O.S.M.I.C.

ACOUSTIC SENSOR

Operating Manual

Team MUTANS - University of Padua
iGEM 2026

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Document information

Scope

This manual comprehensively describes the acoustic sensor developed by Team MUTANS for the noninvasive monitoring of gaseous activity in a benchtop photobioreactor. It is designed to allow for complete and verifiable replication of the hardware, firmware, and software for experimental procedures and data analysis. It can be replicated by a reader with no prior knowledge of electronics, programming, or biology.

Typographical conventions

Symbol	Meaning
[M]	value measured on the real system
[S]	estimated or indicative value (to be confirmed)
IN PROGRESS	activity to be carried out or result not yet available
ERROR	mistake actually made during development, with lesson learned
REQUISITE	non-negotiable operational constraint
code	identifiers, commands, file names, constants

Units follow SI (International System of Units). ADC amplitudes are raw values from the analog-to-digital converter (0–1023 corresponding to 0–5 V). Decibels are relative unless otherwise indicated.

S2S = Sweep-to-Sweep centroid variation

Structure of the manual

• **Part I: System Description**: What it measures, on what physical basis, with what specifications.

• **Part II: Implementation**: Materials, electronics, mechanics, firmware, software, algorithms.

• **Part III: Use**: Safety, maintenance, operating procedures, experimental protocols, analysis, data management.

• **Part IV: Characterization and Validation**: Experimental results, uncertainties, operating regime.

• **Part V: Quality, Compliance, and Development**: Errors, Diagnostics

• **Appendices**: Glossary, Constants. Formul

0. The project in two pages

This manual is long because it is meant to let anyone rebuild and check our sensor down to the last resistor. These two pages are for readers who know nothing about the project: they explain why the sensor exists, what it does, how we tested it and what is still open, and tell you where to go next.

0.1 Why this sensor exists

Our iGEM project, C.O.S.M.I.C., works with cyanobacteria: microscopic organisms that, like plants, use light to produce oxygen. Our goal is to use them, one day, to help supply oxygen to people living on Mars, where the next delivery from Earth could be months away.

A living system that keeps people alive has to be watched, you need to know whether the culture is working. The usual way is to put a probe inside the culture and this has two problems:

- Every probe that goes in is a way in for contamination, for the culture and, in space, for the planet being explored.
- A probe only sees the oxygen dissolved in the water: Water can hold only a little oxygen, about 7–8 milligrams per litre. Once it is full, the extra oxygen comes out as bubbles, like the gas in a fizzy drink and the probe cannot see it.

Can we tell whether a culture is releasing gas bubbles without ever touching it?

0.2 The idea in one picture

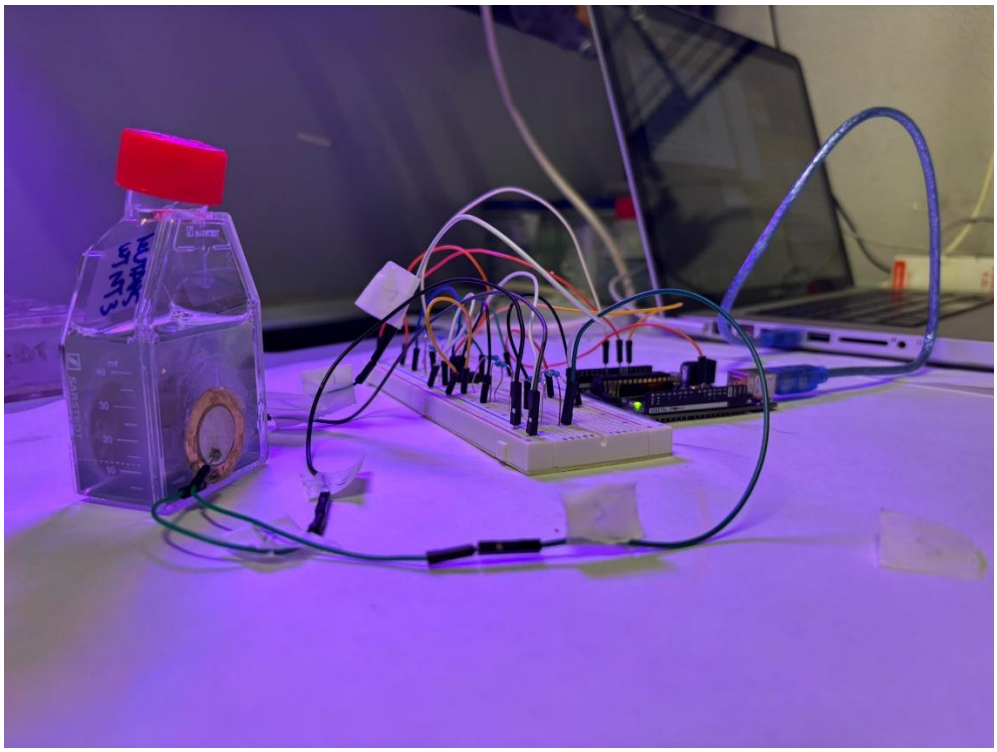


Figure 0: Two piezo discs are clamped on opposite outer walls of a T25 culture flask. One works as a loudspeaker (TX), the other as a microphone (RX). An Arduino drives the first, reads the second through a one-chip amplifier, and sends the data to a laptop, where Python scripts turn them into numbers and plots. Nothing touches the culture. Parts cost about €43.

Think of tapping a glass of still water and a glass of sparkling water: they sound different. Gas bubbles make a liquid much “softer” for sound, so a sound crossing a liquid with bubbles in it comes out changed. Our sensor uses this idea:

1. **Send -** Every ≈10 seconds, one disc plays 60 short tones through the flask, from low to high. We call this a sweep.
2. **Receive -** The disc on the other side picks up how much of each tone got through, and a small amplifier makes the signal strong enough for the Arduino to read.
3. **Reduce -** The software turns each sweep into one number: the average pitch of the sound that got through (we call it the *centroid*, and measure it in hertz, Hz, the unit of pitch).
4. **Compare -** In still water, this number stays almost the same from one sweep to the next. When bubbles cross the space between the discs, it jumps. Our main result, S2S (*sweep-to-sweep*), measures how big these jumps are over a recording of about 15 minutes, called a block.

In short low S2S means quiet water and high S2S means bubbles are crossing.

0.3 What the sensor tells you and what it does not

The sensor is a bubble detector. It answers questions like: are bubbles crossing between the discs and are there more now than before? For example, with the lamp on versus off.

It does not:

- measure oxygen or how much gas the culture produces
- tell you which gas it is
- compare recordings made on different days after the discs have been taken off and put back

0.4 How we know it works

We tested the sensor in steps:

Steps	What we did	What we found
Nothing happening	Only water in the flask	The sensor is stable: S2S stays low, about 2–6 Hz
Gas, no biology	Still water, then sparkling water, without moving the discs	S2S was 6–7 times higher with sparkling water
A living plant	The water plant Elodea, lamp on and off, bubbles filmed and counted	About 3 times the floor at ≈ 11 bubbles/min, back down in the dark; at the floor at ≈ 4 bubbles/min. One plant, one session, and the positive block fails our quality gate (Ch. 23.2)
Our cyanobacteria	Cultures under the lamp on a shaker	Not yet conclusive: each session removed one source of noise (lamp switching, then the shaker), and a follow-up test traced the remaining electrical noise to the laptop charger (Ch. 23.5–23.7)

Every comparison is made inside one session, without moving the discs. Before any result is read, each block must pass a quality gate with fixed thresholds that checks the contact between discs and flask and the background noise (Ch. 14.6). A block that fails is reported, but never counted as positive or negative.

0.5 What is proven and what is still open

Proven

- The sensor is stable when nothing happens, and clearly detects free gas (sparkling water, 6–7 times).
- It flags its own bad recordings.
- It costs about €43 in parts, runs only on USB power and never touches the culture.

Still open

- Detecting the bubbles of our cyanobacteria.
- Comparing recordings across different days.
- A test with bubbles produced at known rates, to say precisely how well it works.

0.6 How to read this manual

If you want to…	Read
Understand what is measured and why it works	Part I (Ch. 1–6)
Build the sensor	Part II (Ch. 7–13)
Run an experiment	Part III (Ch. 15–20) and the User Guide
Judge how far the results go	Part IV (Ch. 21–24)
Learn from our mistakes	Part V (Ch. 25–27)

PART I: SYSTEM DESCRIPTION

1. Technical summary

The system is an active acoustic sensor in a through-transmission configuration for benchtop photobioreactors. Two piezoelectric disk transducers (Ø 27 mm) are attached externally, on opposite walls, to a Sarstedt T25 culture flask. The transmitting transducer (TX), driven by an Arduino Uno, emits a 60-tone sweep on a logarithmic scale between 1000 and 3500 Hz, the receiving transducer (RX) collects the signal after passing through the culture. A single operational amplifier (LM358) analog front-end amplifies and centers the signal, which is digitized by the microcontroller's 10-bit converter and transmitted to the computer via serial link.

A Python software chain captures, segments, and reduces each sweep to a set of spectral descriptors; the primary metric is the standard deviation of the spectral centroid differences between consecutive sweeps (sweep-to-sweep centroid variation), computed by 03_features.py together with the coupling gate (Ch. 14.6), which responds to the passage of gas bubbles through the acoustic beam and is insensitive to slow coupling drifts.

The measurand is gaseous activity. It is defined as the presence and dynamics of gas bubbles in the acoustic beam. In a photobioreactor these are attributed to photosynthetic O_2 released once the medium is supersaturated and is an attribution based on the experimental design and not on acoustics (Ch. 2.3). The system does not measure dissolved oxygen, does not identify the chemical species of the gas, and does not count bubbles. It detects fluctuations in the fraction of gas in the beam, with a response that scales with the volume and size of the bubbles.

What's new: The literature on the acoustics of photosynthetic organisms employs passive hydrophone methods, which listen to the noise produced by bubbles (Freeman et al., PLoS ONE 2018; Kratochvil & Pollirer, Sci. Rep. 2017; PAM monitoring of seagrass meadows, 2025). The present system is active, in fact it transmits a controlled acoustic sweep through the culture and measures its perturbation (through-transmission). To our knowledge, this configuration has not previously been applied to monitoring photosynthetic bubbles in photobioreactors.

2. Principle of measurement and definition of the measurand

2.1 The Measurand

The measurand is the gaseous activity in the volume traversed by the acoustic beam, defined as the presence, relative intensity, and temporal dynamics of free gas bubbles. Operationally, the system returns:

1. a primary activity quantity (Sweep to sweep, S2S, in Hz) sensitive to fluctuations in the gas fraction

2. a coupling state quantity (peak amplitude, in ADC) used as an index of measurement quality

3. secondary quantities (absolute energy, centroid, high-frequency ratio) for intra-session comparisons

2.2 Principle

An acoustic wave passing through a liquid containing gas bubbles experiences three effects: a reduction in propagation speed, attenuation, and diffusion, with frequency dependence related to the size of the bubbles. This results in a perturbation of the transmission spectrum of the piezo-wall-liquid-wall-piezo system. The system measures this spectrum repeatedly (one sweep every ≈10 s; ≈5 s of sweep followed by 5 s of passive listening) and quantifies how much it changes from one sweep to the next.

2.3 Declared scope

The system:

• detects the presence of gaseous activity (bubbles passing through the beam).

• estimates its relative intensity within the same measurement session.

• monitors its temporal dynamics and response to stimuli (for example light ON/OFF).

The system does not:

• measure dissolved oxygen (the dissolved phase is acoustically invisible at any saturation level).

• provide absolute production values (mg/L, mL O_2/min) without external calibration.

• identify the chemical species of the gas: the assignment to O_2 is based on the experimental design (light dependence, reversibility, absence in controls), not on acoustics.

• count bubbles: the response scales with volume and size, not with number.

• have validated performance in containers other than the T25 flask or at larger volumes.

3. Measurement model

3.1 Fate of the oxygen produced

The oxygen produced by photosynthesis follows two phases:

1. **Dissolved phase**: Until saturation, the O_2 dissolves in the medium. The dissolved gas does not form bubbles, does not appreciably alter acoustic propagation in the operating band, and is therefore invisible to the sensor.

2. **Gaseous phase:** Beyond saturation, the excess O_2 nucleates into bubbles that rise and leave the liquid. This phase is the one detected by the sensor.

3.2 Complementarity with the dissolved oxygen probe

A dissolved oxygen probe (galvanic, polarographic, or optical) measures the partial pressure of O_2 in solution and reads correctly even above 100% saturation. In active photosynthesis, readings of 150–250% are common. What the probe cannot see is the oxygen leaving the solution as bubbles. Above saturation, production is split between a portion that remains dissolved (tracked by the probe) and a portion that vents as gas (invisible to the probe). When bubbling balances production, the dissolved concentration settles on a supersaturated plateau and its time derivative (the usual estimator of the production rate) approaches zero while production continues. In this regime, the probe underestimates the actual production by the fraction vented as a gas, which it cannot measure.

Regime	DO probe	Acoustic sensor
Sub-saturated	measures concentration and rate (slope)	no signal
Oversaturated, bubbling incipient	measures concentration; the slope rate begins to underestimate	detects intermittent bubbles
Supersaturated, bubbling established	plateau concentration; uninformative slope rate	detects gas flow (in intermittent mode)

The two instruments are therefore complementary: the probe covers the dissolved phase, the acoustic sensor the gaseous phase. The probe remains useful even above saturation (supersaturation level; production estimation via gas-liquid exchange model); the acoustic sensor's value lies in the direct detection of the gaseous phase, without vent modeling, externally and non-invasively.

3.3 Assumptions

ID	Statement	State
A1	O_2 production is proportional to photosynthetic activity	Consolidated biology
A2	Above saturation, the bubble flux increases with the production rate	plausible; not quantitatively calibrated
A3	The acoustic signature varies measurably and repeatably with the bubbles	Demonstrated
A4	The Sweep-to-Sweep centroid variation (S2S) follows the flow of intermittent bubbles	Demonstrated (one block of Trial 5; r = 0.91 over six blocks, driven by that block, Ch. 23.2)
A5	Intra-session comparisons erase systematic pairing errors	Consistent with all data
A6	The Sweep-to-Sweep centroid variation (S2S) is blind to steady state bubbling	Demonstrated: declared limit

4. Physical foundations

4.1 Minnaert resonance

A gas bubble in a liquid behaves like an oscillator: the gas is the spring, the surrounding liquid is the mass. For air in water at atmospheric pressure, the resonant frequency is approximated by

$$f = \frac{1}{2\pi a}\left(\frac{3\gamma\, p_A}{\rho}\right)^{1/2}.$$

with “a” bubble radius. In the operating band 1000–3500 Hz, bubbles with a radius approximately 0.9–3.3 mm, i.e. with a diameter approximately 2–7 mm, resonate.

Implications for photosynthetic bubbles: Bubbles produced by microbial cultures are typically submillimeter; their resonance falls at 13–33 kHz, out of range. They do not benefit from any resonant amplification and contribute only their volume to the signal. Many small bubbles can therefore

produce a lower signal than a few large bubbles. This finding, which emerged in Trial 8 (08_elodea_4), influences the Anabaena protocol.

4.2 Wood's equation

In a liquid–gas mixture the speed of sound is given by

$$c = \frac{1}{\sqrt{\rho_m \left(\frac{\alpha}{P} + \frac{1-\alpha}{K_l} \right)}}$$

where α is the volume fraction of gas. Since the mixture has the density of a liquid but a compressibility dominated by the gas, a gas fraction of 10^{-3} lowers the speed of sound from ≈ 1480 m/s to ≈ 300–360 m/s (4–5×, isothermal to adiabatic gas), and 10^{-2} to ≈ 100 m/s. Wood's equation holds below bubble resonance, which is the case here (Ch. 4.1)

The system is therefore very sensitive to small amounts of free gas in the beam.

4.3 Attenuation and diffusion

The bubbles absorb and scatter acoustic energy, with a maximum near their resonance.

Observed effects [M]: reduction of received energy (-3.81 dB with sparkling water) and peak amplitude (-40%).

4.4 Spectrum of the loaded system

The peaks observed in the transmission spectrum (1550/1583 Hz, 2120 Hz, 2454/2558/2612 Hz) are resonances of the loaded system (transducers, polystyrene walls, liquid column) and not bubble resonances. The free piezoelectric disk resonates around 4 kHz; when loaded by the wall and the liquid, the resonance drops and becomes structured. The information about the bubbles lies in the perturbation of this spectrum between consecutive sweeps, not in its absolute position. At 1–3.5 kHz the wavelength in water (≈ 0.4–1.5 m) is much larger than the flask, so the transducers excite standing modes of the whole flask–liquid system rather than a collimated beam. In this manual, “acoustic beam” denotes the liquid region between the discs, and the sensitivity to a bubble may depend on where it crosses these modes.

4.5 Rationale for the high-pass metric

Let c(t) be the spectral centroid measured once for sweep (Δt ≈ 10 s).

• A slow drift (coupling settling, temperature) shifts c(t) by tens of Hz over a period of minutes; the difference between consecutive sweeps is small and almost constant.

• The transit of a bubble produces a c(t) excursion that develops within one or two sweeps (tens of Hz with sparkling water, Ch. 22): what separates it from drift is the time scale, not the size.

The standard deviation of consecutive differences, $S2S = std(c_{i+1} - c_i)$, attenuates the first and preserves the second: it acts as a high-pass filter on the centroid. It is the primary metric of the system.

Limiting consequence: A continuous, stationary flow of bubbles shifts the centroid to a new mean value and maintains it there; the consecutive differences become small again, and the S2S does not

increase, even in the presence of gas. This behavior has been observed experimentally and is treated as an intrinsic limit.

5. Technical specifications

Parameter	Value	Origin
Configuration	through-transmission, external transducers, opposed	Project
Transducers	2 piezoelectric disks, Ø 27 mm (brass/PZT)	Project
Excitation bandwidth	1000-3500 Hz nominal, 60 logarithmic steps. Generated grid ≈ 999–3427 Hz (the last step is F_MIN·(F_MAX/F_MIN)^(59/60))	Firmware
Analysis bandwidth	1000–3200 Hz	Characterization [M]
Sweep cadence	≈ 10 s (≈ 5 s sweep + 5 s passive listening) [M]	Firmware
ADC resolution	10 bits (0-1023 on 0-5 V)	Arduino Uno
Front-end gain	≈ 5 (500 kΩ / 100 kΩ), bias 2.5 V	Project
Typical peak amplitude (coupled)	130-200 ADC	[M]
Background (dry transducers)	≈ 10 ADC	[M]
Intra-run centroid repeatability	σ ≈ 2-4 Hz	[M]
Inter-run reproducibility (remounting)	up to ≈ 281 Hz deviation	[M]
S2S floor (no bubbles)	2.4-5.8 Hz with good coupling. Can go up to ≈ 8 Hz with degraded coupling (Ch. 21.3)	[M]
Detection limit (intermittent bubbles)	between ≈ 4 and ≈ 11 bubbles/min; one session (Trial 5)	[M]
Observed biological response time	≈ 30 s from light on	[M]
Interface	USB serial, 115 200 baud, CSV	Firmware/Software
Power supply	5 V from USB	
Validated vessel	Sarstedt T25, Polystyrene	
Component cost	42,72€	BOM

6. Architecture

Level	Elements	Function
Hardware	Transducers, front end, microcontroller, mechanical support, coupling	Stimulus generation, conditioning, and digitization
Firmware	Acoustic_Sensor.ino	State machine: SWEEP / BURST / NOISE / RAW
Software	Four Python scripts	Acquisition, analysis, metrics, monitoring

PART II: IMPLEMENTATION

7. Bill of Materials (BOM)

7.1 Electronics

All prices listed below are from Amazon to ensure accessibility. Please note that prices may vary when purchasing from specialized retailers. We used an Arduino Uno kit to demonstrate the accessibility and low cost of the components used, so all the pieces with * in the cost column are included in the kit. Each ELEGOO UNO R3 Super Starter Kit contains the components marked, the cost of the one we used is 37.35 €.

#	Component	Specifications	Amount	Price [€]	Note
E1	Arduino Uno R3 or compatible clone	ATmega328P, 16 MHz, 10-bit ADC	1	*	Clones with CH340 converter require the corresponding driver
E2	Piezoelectric disk	Ø 27 mm, brass substrate, PZT, two-wire	2 (+3 spare)	15,59 (for 20 pieces)	fragile solder joints
E3	LM358 operational amplifier	DIP-8	1 (+2)	3,81	
E4	Ceramic capacitor	100 nF	1	*	AC input coupling
E5	Resistor	100 kΩ	1	*	Input
E6	Resistor	1 MΩ	2	*	in parallel: 500 kΩ feedback
E7	Resistor	10 kΩ	3	*	Two for the bias divider, one for protection on A0
E8	Resistor	100 Ω	1	*	In series with the TX
E9	Breadboard and jumpers		1	*	Later, a stripboard or PCB
E10	USB A-B cable		1	*	

Total cost: 42.72 € (kit + LM358 + the two piezo discs actually used, pro-rated from the pack of 20)

7.2 Mechanics and coupling

#	Component	Specification	Qty	Notes
M1	Sarstedt T25 flask	Polystyrene, flat walls, 25 cm^2	>= 1	Reference container for the entire characterization
M2	3D-printed stand	porta_piezo_T25.scad -> STL, PLA, or PETG	1	

#	Component	Specification	Qty	Notes
M3	M3 screws with nuts	20–25 mm	2–4	Tightening
M4	Vaseline		1	Coupling. Vaseline only for short sessions
M5	Isopropyl alcohol, paper towels			Cleaning of transducers and electronics only. On the polystyrene flask wall use 70 % ethanol without rubbing (Ch. 15.2)

7.3 Material for experimental protocols

#	Components	Use
P1	Bottled sparkling water	Abiotic positive control
P2	Straw	Coarse bubble abiotic positive control
P3	Elodea canadensis	Biological validation with plants (Ch. 18.2)
P4	LED	Photosynthetic illumination, 100–300 µmol m^{-2} s^{-1} [S]
P5	Box or opaque cloth	Dark blocks
P6	Smartphone or webcam with tripod	Video reference (Ch. 18.4)
P7	$NaHCO_3$	Carbon source (≈ 10 mM)
P8	Nostoc (Anabaena) sp. PCC 7120 culture in BG-11, spectrophotometer (OD_{750})	Biological validation in a photobioreactor
P9	Lamp on a separate mains outlet	Light/dark blocks. Keep lamp and cables away from the RX wiring (Ch. 14.6)
P10	Thermometer	Water temperature <=25 C
P11	Orbital mixer / shaker (optional)	To mix the culture between blocks only, off during recordings

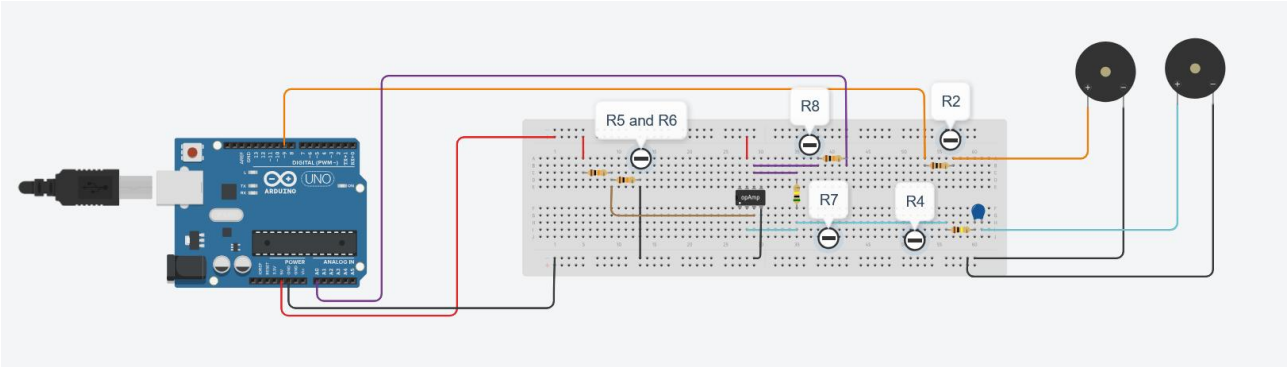
8. Electronic circuit: design, assembly and testing

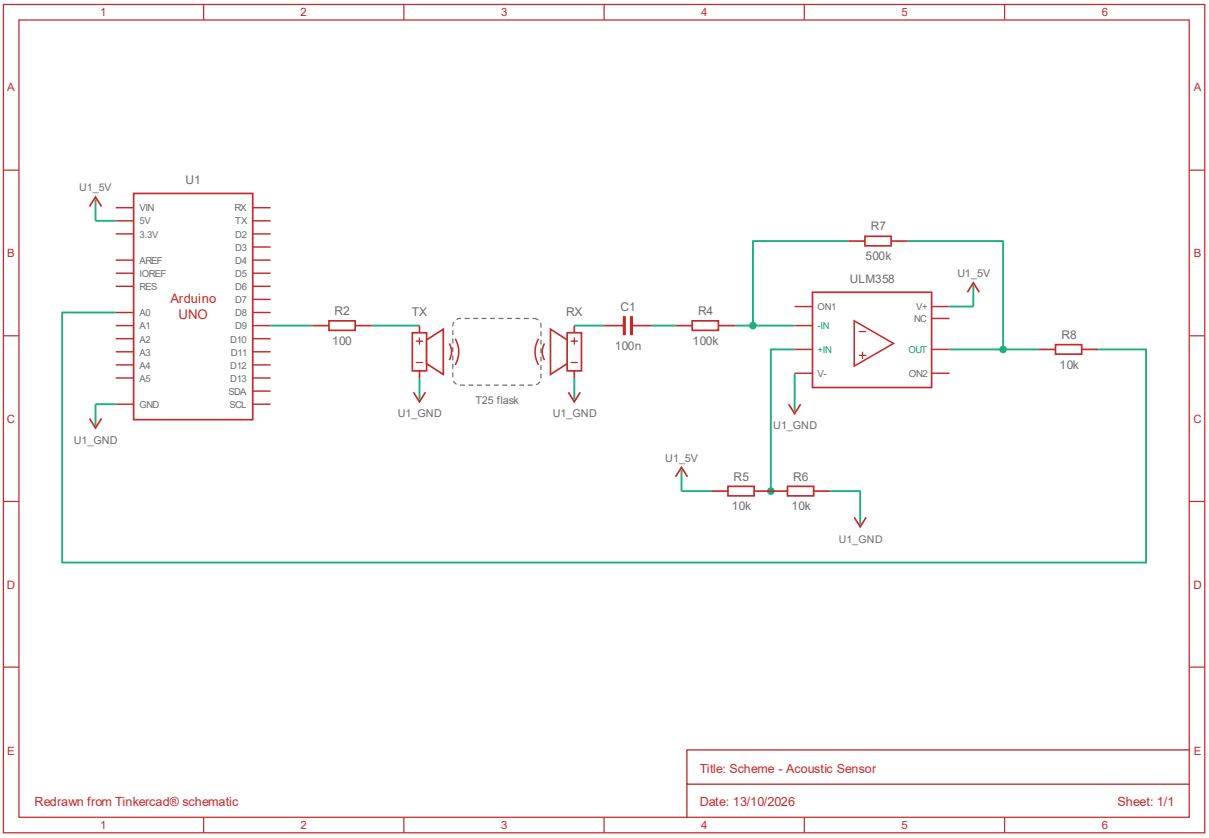
8.1 Front-end requirements

The RX transducer generates a millivolt amplitude signal, bipolar with respect to 0 V. The Arduino converter accepts 0–5 V. The front-end must therefore (i) amplify and (ii) translate the signal around 2.5 V, so that the positive and negative half-waves remain within the conversion range.

8.2 Scheme

Single op-amp inverting configuration with half-supply reference.





Element	Value
C1	100 nF
R2	100 Ω
R4	100 kΩ
R5, R6, R8	10kΩ
R7	500 kΩ (2 × 1 MΩ in parallel, the nearest single standard value)
ULM358	Operational amplifier LM358
TX, TR	Piezoelectric transducer
Arduino Uno R3	Microcontroller board

Pin 9 generates a square wave at the requested frequency via tone(); the resistor limits the peak current on the capacitive load.

8.3 Breadboard assembly procedure

1. Insert the LM358 across the central groove, with the reference notch oriented; pin 1 is adjacent to the notch.

2. Connect 5 V to pin 8 and GND to pin 4.

3. Build the bias divider (10 kΩ from 5 V to a free node, 10 kΩ from the node to GND) and connect the node to pin 3.

4. Insert the two 1 MΩ resistors (or one 500 kΩ resistor) in parallel between pin 2 and pin 1.

5. Connect the RX piezo signal wire to the 100 nF capacitor, then to the 100 kΩ resistor, then to pin 2.

6. Connect pin 1 to the 10 kΩ resistor, then to A0.

7. Connect pin 9 to the 100 Ω resistor, then to the TX piezo signal wire.

8. Connect the second wires of both transducers to GND.

9. Keep the connections short; secure the transducer wires mechanically (hot glue or tape) to relieve the solder joints of stress.

8.4 Electrical check (multimeter, before connecting to the computer)

Measurement	**Point**	**Expected value**	**Interpretation in case of deviation**
DC Voltage	Pin 3 -> GND	2.5 V ± 0.2 V	missing or incorrect bias (E-01)
DC Voltage	Pin 1 -> GND, idle	≈ 2.5 V	0 V or ≥ 3.5 V: op-amp saturated (on a 5 V supply the LM358 output cannot exceed ≈ Vcc - 1.5 V ≈ 3.5 V)
Resistance	Pin 1 -> A0	10 kΩ	lower value: parasitic divider (E-02)
Resistance	Pin 2 -> Pin 1	≈ 500 kΩ	
Continuity	GND Arduino -> Pin 4 -> Piezo cold leads	Continuity	Mandatory common ground

8.5 Functional check

1. Load the firmware; open the serial monitor at 115 200 baud: # READY should appear.

2. Send S: freq, amplitude, ms, S lines should scroll.

3. Dry, uncoupled transducers: amplitudes ≈ 10 ADC (background). Expected behavior.

4. Transducers mounted on the flask with couplant: peak of 130–200 ADC in the 1200–1600 Hz region. If the peak remains ≈ 10 ADC: no coupling (E-03).

5. Amplitude constant at a high value on every step. Flat-topped spectrum, with the LM358 on 5 V the peak-to-peak cannot exceed ≈ 3.5 V ≈ 700–750 ADC [S].

6. Interference check (REQUISITE before every biological session): you need a flask filled with water and all the auxiliary equipment in its recording configuration with the laptop on battery. Record >= 10 min in sweep mode while switching the lamp (and any agitator) OFF and ON. Acceptance: passive

noise (std of P rows) ≤ 2 ADC in both states, and no step in rms_abs_db or in the > 3200 Hz bins at the transitions.

9. 3D printed mechanical support

9.1 Motivation

The dominant source of error in the system is repositioning: manually reassembling the transducers between sessions shifts the loaded resonance up to ≈ 281 Hz [M]. A mount that constrains position and pressure in a repeatable way attacks the cause of the error rather than the symptom.

9.2 Design requirements

- Hold TX and RX coaxially on opposite walls of the T25 flask, at a fixed height that the bubble path crosses.
- Seat the flask against a registration lip, so that it returns to the same position after cleaning.
- Hold each Ø 27 mm disc in a pocket of defined clearance, with the wires routed through a strain relief.
- Apply a repeatable clamping force: M3 screws tightened to contact plus 1/4 turn. Over-tightening squeezes out the couplant.
- Leave the region between the transducers visible to the camera (Ch. 18.4).
- Material: PLA or PETG but PETG tolerates the cleaning agents better (Ch. 15.2).

9.3 Fabrication and assembly

1. Check the fit of the flask against the lip and of the discs in their pockets before applying the couplant.

2. Apply the couplant (Ch. 10), insert the discs and seat the flask against the lip.

3. Tighten the screws to contact, then a further ¼ turn, alternating between them.

10. Acoustic coupling

10.1 Principle

A dry transducer is separated from the wall by a microlayer of air that reflects almost all of the acoustic energy due to impedance mismatch. The couplant fills the microlayer. Measured effect [M]: peak amplitude from ≈ 11 ADC (dry) to ≈ 150 ADC (coupled), with background ≈ 10 ADC. In the absence of couplant the instrument is inoperative.

10.2 ERROR E-07: Cold flow of Vaseline

Vaseline (petrolatum) provides excellent coupling in the first few minutes, but under clamp pressure it progressively flows out of the interface (cold-flow). Dedicated measurement [M]: peak amplitude −42% in ≈ 28 minutes, with an almost monotonic trend. In Trial 6, degradation (−35% from the baseline block) made the measurement uninterpretable despite the presence of 20–30 bubbles/min documented on the video. Tighter clamping worsens the phenomenon.

10.3 Resolution of the problem

- Vaseline without support is for sessions less than 20 minutes after warm-up.
- Alternative to couplant: silicone grease (for gaskets, not sealants) or ultrasound gel, both of which are low-creep. For characterization: 30-min post-plateau testing. The acceptance criterion is a stable peak amplitude within +-5%.
- Thin, uniform layer on a clean, dry surface.
- Moderate, repeatable pressure (recording lip, stop screws + 1/4 turn).
- No recording before the peak amplitude plateaus.

11. Firmware

11.1 Constants

Constant	Value	Meaning
F_MIN	1000 Hz (first observed step ≈ 999 Hz)	Start of sweep
F_MAX	3500 Hz	Nominal end of sweep (reduced from 20,000, Ch. 11.4). The last step actually generated is ≈ 3427 Hz
N_STEPS	60	Steps per sweep, logarithmic grid
Settle time (hard-coded delay(40), it is not a named constant)	40 ms	Mechanical stabilization per step
DWELL_MS	80 ms	Total time per step (settle + measurement + wait)
sweep_interval	5000 ms	Passive listening between sweeps (P rows every ≈ 100 ms)
Measurement window	>= 3 periods, limited to 1500–4000 µs	Peak-to-peak measurement
Baud rate	115 200	Serial
Sweep duration	(40 + ≈ 3 + 40) ms + serial output ≈ 5 s	One metric point every ≈ 10 s [M]

The logarithmic scale concentrates the steps at low frequencies, where the main resonance and most of the information reside.

11.2 State machine

Command	Mode	Behavior	Data rows
S	SWEEP	Repeated sweeps: passive listening during the 5 s pause between sweeps (one P row every ≈ 100 ms)	f,ampl,ms,S e 0,ampl,ms,P
B	BURST	Continuous fixed tone (BURST_FREQ = 8000 Hz), amplitude every 50 ms	f,ampl,ms,B
N	NOISE	TX off, listening only (loop ≈ 2.5 kHz; rows are skipped when the serial buffer is full, so the logged rate is lower [S])	0,ampl,ms,N

Command	Mode	Behavior	Data rows
R	RAW	Repeated blocks of 512 raw ADC samples between # RAW_START and # RAW_END, one block every ≈ 200 ms	0,val,ms,R
X	STOP	Switches TX off and returns to idle, then send S to restart. 01_data_from_sensor.py sends X when it stops	

11.3 Measurement of amplitude

For each frequency the firmware calculates the period, opens a window ≥ 3 periods (limited to 1500–4000 µs), samples the ADC continuously while maintaining the maximum and minimum, and returns the peak-to-peak amplitude vmax − vmin (0–1023). This is the primary physical quantity.

ERROR E-12: Average instead of peak-to-peak. The first version calculated the average of the samples. A waveform centered at 2.5 V has an average ≈ 512 regardless of the amplitude: the result was constant and the graphs were flat, with plausible but uninformative data. The error does not generate obvious malfunctions and is therefore particularly insidious. Dry test ≈ 10 ADC, coupled > 100 ADC.

11.4 F_MAX Choice

The characterization showed that the entire information content is below ≈ 3.2 kHz. Reducing F_MAX from 20,000 to 3500 Hz, with the same 60 steps, increases the resolution in the useful band by ≈ 2.4×: the main resonance, previously represented by a single point, is resolved as a 1550/1583 Hz doublet, and the fine structure 2454/2558/2612 Hz appears.

11.5 Reference implementation

```
/*
ACOUSTIC BIOREACTOR - PROGRAM FOR ARDUINO UNO R3
Trasmission of the sweep and acquisition of the data

CONNECTION
Piezoelectric sensor - Trasmission  -> Pin 9 (PWM)
Piezoelectric sensor - Receiver    -> Pin A0 (ADC)
Led status                  -> Pin 13 (built-in) (FOR DEBUG)

SERIAL PROTOCOL (to send data to python)
Every lines has: "F,AMPL,TIMESTAMP,MODE\n"
F       = frequency
AMPL    = peak-to-peak amplitude ADC (vmax - vmin, from 0 to 1023) measured over a
          window of at least 3 periods of the tone (NOT an average of N samples)
TIMESTAMP = millis() of Arduino
MODE    = S(sweep)/P(passive)/B(burst)/N(noise)

MODALITY
1. SWEEP  -> scan from F_MIN to F_MAX in N_STEPS steps
2. BURST  -> fixed frequency continuous tone, log raw ADC
3. NOISE  -> TX silence, listen only (noise baseline)

Modality is selected by arduino by Python from serial route
*/

// PIN
#define PIN_TX     9   // exit PWM (= Pulse Width Modulation) to Piezoelectric sensor - trasmission
```

```
#define PIN_RX    A0   // input ADC piezoelectric sensor - receiver
#define PIN_LED   13   // Control LED

// SWEEP PARAMETERS
// F_MAX = 3500 Hz: measured useful band is 1-3 kHz (dominant loaded resonance is
// ~1550/1583 Hz, secondaries ~2120 and ~2454/2558/2612 Hz); above ~3 kHz there is
// only noise floor (~10 ADC),
// so scanning up to 20 kHz was wasted time. We are concentrating the same 60 log steps in
// 1-3.5 kHz raises frequency resolution ~4x where the signal actually is.
#define F_MIN     1000  // Hz - minimum frequency
#define F_MAX     3500  // Hz - maximum frequency (useful band)
// Wide-band diagnostic mode: uncomment to re-characterize the system (PWM Uno limit ~31kHz)
// #define F_MAX   20000
#define N_STEPS     60  // number of frequency steps
#define DWELL_MS    80  // ms for every frequency (stabilization + plotting)
#define N_AVG       32  // legacy: not used, measure_amplitude() works on a timed peak-to-peak window

//  BURST PARAMETERS
#define BURST_FREQ  8000 // Hz - frequency burst (default)
#define BURST_MS    500 // time burst (after there is nothing, after there is burst, ...)

// BAUD RATE (For Arduino)
#define BAUD      115200

// GLOBAL STATE
char mode      = 'I';  // current modality: I(idle)/S/B/N/R - starts idle, waits for Python's start command
bool tx_active  = false; // Trasmission (TX)
unsigned long sweep_interval = 5000; // ms beetween sweep


void setup() {
 Serial.begin(BAUD);
 pinMode(PIN_TX,  OUTPUT);
 pinMode(PIN_LED, OUTPUT);

 // ADC: prescaler at 16 -> ~77kHz sample rate (default 125kHz/10bit)
 // made samples more faster for signals with high frequency
 ADCSRA = (ADCSRA & 0xF8) | 0x04; // prescaler 16

 // discard first 5 samples, because the first sample is always higher than the others, it is a systematic error of the ADC
 for (int i = 0; i < 5; i++) analogRead(PIN_RX);

 // waiting for power supply stabilization
 delay(200);

 //For the problem of the first tone, send a short tone at F_MIN
 tone(PIN_TX, F_MIN);
 delay(150);
 noTone(PIN_TX);

 // header CSV for Python
 Serial.println("# BIOREATTORE ACUSTICO v5.1");
 Serial.println("# freq_hz,ampl_adc,timestamp_ms,mode");
 Serial.println("# READY");

 digitalWrite(PIN_LED, HIGH);
}


void loop() {

 // Arduino see if python has send some command, always
 if (Serial.available()) {
  char cmd = Serial.read();
  parse_command(cmd);
 }

 // selector of state, state machine
```

```
  switch (mode) {
    case 'I': /* idle: TX off, do nothing until Python sends a command */ break;
    case 'S': run_sweep();  break;
    case 'B': run_burst();  break;
    case 'N': run_noise();  break;
    case 'R': run_raw();    break;
  }
}


// decode of the instruction send by python

void parse_command(char cmd) {
  switch (cmd) {
    case 'S':
      mode = 'S';
      Serial.println("# MODE:SWEEP");
      break;
    case 'B':
      mode = 'B';
      Serial.println("# MODE:BURST");
      break;
    case 'N':
      mode = 'N';
      tx_off();
      Serial.println("# MODE:NOISE");
      break;
    case 'R':
      mode = 'R';
      Serial.println("# MODE:RAW");
      break;
    case 'X':
      // total stop of trasmission TX
     mode = 'I';
     tx_off();
        Serial.println("# TX OFF");
        break;
    }
  }


  // MODALITY 1: FREQUENCY SWEEP
  // Generate a frequency, wait for stabilization, measure, advance

  void run_sweep() {

    // logarithmic step calculation (is more useful than linear calculation for acoustics)
    // f[i] = F_MIN * (F_MAX/F_MIN)^(i/N_STEPS)
    float log_min = log((float)F_MIN);
    float log_max = log((float)F_MAX);
    float log_step = (log_max - log_min) / N_STEPS;

    // marker start sweep for Python
    Serial.print("# SWEEP_START,");
    Serial.println(millis());

    for (int i = 0; i < N_STEPS; i++) {

      // if there are interruption by Python, interrupt all the process
      if (Serial.available()) {
        char cmd = Serial.read();
        if (cmd == 'X' || cmd == 'N') {
          parse_command(cmd);
          return; // stop sweep
        }
      }

      // calculate current frequency (log scale)
```

```
    float f = exp(log_min + log_step * i);
    unsigned int freq = (unsigned int)f;

    // generation on the piezoelectric sensor TX
    tone(PIN_TX, freq);
    tx_active = true;

    // mechanical stabilization of the piezoelectric sensor
    // first 40ms are skipped
    delay(40);

    // measure the peak-to-peak amplitude of the signal at the current frequency
    int ampl = measure_amplitude(freq);

    // waiting for the dwell
    delay(DWELL_MS - 40);

    // output CSV: freq,ampl,timestamp,mode
    Serial.print(freq);
    Serial.print(',');
    Serial.print(ampl);
    Serial.print(',');
    Serial.print(millis());
    Serial.print(',');
    Serial.println('S');
  }

  // end sweep - silence TX
  tx_off();

  Serial.print("# SWEEP_END,");
  Serial.println(millis());

  // Pause between sweeps
  // During the pause, listen passively (useful for bubble noise)
  unsigned long t0 = millis();
  while (millis() - t0 < sweep_interval) {
    if (Serial.available()) {
      parse_command(Serial.read());
      return;
    }
    // passive sampling every 100ms
    int passive = analogRead(PIN_RX);
    Serial.print(0);      // freq=0 for passive hearing
    Serial.print(',');
    Serial.print(passive);
    Serial.print(',');
    Serial.print(millis());
    Serial.print(',');
    Serial.println('P');   // P = Passive
    delay(100);
  }
}


// MODALITY 2: BURST
// continuously BURST_FREQ, log peak-to-peak ADC every 50ms
// Good for the monitoration in real time at static freq

void run_burst() {
  tone(PIN_TX, BURST_FREQ);
  tx_active = true;

  unsigned long t0 = millis();

  while (mode == 'B') {

    if (Serial.available()) {
```

```
    parse_command(Serial.read());
    break;
   }

   //peak-to-peak over a window of at least 3 periods
   int ampl = measure_amplitude(BURST_FREQ);

   Serial.print(BURST_FREQ);
   Serial.print(',');
   Serial.print(ampl);
   Serial.print(',');
   Serial.print(millis());
   Serial.print(',');
   Serial.println('B');

   delay(50); // 20 Hz log rate
  }
}


// MODALITY 3: NOISE (pure listening, TX off)
// Captures passive acoustic noise from the reactor
// Includes: bubble noise, pump, background

void run_noise() {
 tx_off();

 while (mode == 'N') {

  if (Serial.available()) {
   parse_command(Serial.read());
   break;
  }

  int sample = analogRead(PIN_RX);
  if (Serial.availableForWrite() > 20) {
   Serial.print(0);
   Serial.print(',');
   Serial.print(sample);
   Serial.print(',');
   Serial.print(millis());
   Serial.print(',');
   Serial.println('N');
  }
  delayMicroseconds(400); // ~2.5kHz sample rate
 }
}


// MODALITY 4: RAW  raw ADC dump at full speed
// Only for short acquisitions (serial buffer becomes saturated)
// In Python: read with read_until or circular buffer

void run_raw() {
 tx_off();

 // send 512 sample raw, then wait
 Serial.println("# RAW_START");
 for (int i = 0; i < 512; i++) {
  Serial.println(analogRead(PIN_RX));
 }
 Serial.println("# RAW_END");

 delay(200); // pause between raw bursts
}

// Measure the peak-to-peak amplitude (vmax - vmin, 0..1023) of the signal at a given frequency
// The window must be long enough to capture several periods of the signal, otherwise the amplitude will be underestimated
```

```
// it is necessary because the piezoelectric sensor has a mechanical inertia and the signal is not immediately stable:
// at 1kHz 3 periods are about 3ms, at 20kHz they are about 0.15ms
int measure_amplitude(unsigned int freq) {
  unsigned long period_us = (freq > 0) ? (1000000UL / freq) : 1000UL;
  unsigned long window_us = 3UL * period_us;  // the minimum window is 3 periods of the signal, to capture the amplitude correctly
  if (window_us < 1500UL) window_us = 1500UL;  // minimum: many samples even at high frequencies
  if (window_us > 4000UL) window_us = 4000UL;  // ceiling: don't slow down too much at 1 kHz

  int vmax = 0;
  int vmin = 1023;
  unsigned long t_start = micros();
  while ((unsigned long)(micros() - t_start) < window_us) {
    int v = analogRead(PIN_RX);
    if (v > vmax) vmax = v;
    if (v < vmin) vmin = v;
  }
  return vmax - vmin;   // peak-to-peak amplitude, 0..1023
}

// UTILITY: TX off

void tx_off() {
  noTone(PIN_TX);
  digitalWrite(PIN_TX, LOW);
  tx_active = false;
  digitalWrite(PIN_LED, LOW);
  delay(10);
  digitalWrite(PIN_LED, HIGH);
}
```

11.6 Loading the firmware

1. Install the Arduino IDE (arduino.cc -> Software). For clones with a CH340 converter, install the corresponding driver.

2. Connect the board via USB. In Tools -> Board, select "Arduino Uno", then in Tools -> Port, select the displayed port.

3. Open Acoustic_Sensor.ino and press Upload. Wait for the "Done uploading" message.

4. Open the serial monitor at 115,200 baud with a "Newline" line terminator. Verify that there is # READY and then send S and verify data flow.

5. Close the serial monitor before starting Python scripts: the port can only be open by one process at a time. (E-16)

12. Serial interface and data format

12.1 Data lines (Arduino -> PC)

Format: F, AMPL, TIMESTAMP, MODE.

Format	Contents
F	Frequency in Hz; 0 for passive, noise, and raw listening
AMPL	ADC peak-to-peak amplitude, 0–1023
TIMESTAMP	milliseconds (reset at each reset)
MODE	S sweep · P passive between sweeps · B burst · N noise · R raw

RAW mode lines do not follow this format: run_raw() prints a single ADC reading per line between the #RAW_START and #RAW_END markers. Since logger 01 only accepts four-field lines, RAW lines are

discarded during acquisition: RAW mode is intended for direct capture from a serial terminal or for a future FFT, not for the CSV pipeline.

12.2 CSV file format (logger output)

Column	Contents
pc_timestamp	Unix PC time (s, decimal) - alignment key with video, light, and annotations
freq_hz	generated frequency
ampl_adc	ADC amplitude
arduino_ms	millis()
mode	S/P/B/N/R/META
raw_line	original line (diagnostics)

Example:

pc_timestamp,freq_hz,ampl_adc,arduino_ms,mode,raw_line
1784304724.86,0,0,0,META,# ACOUSTIC BIOREACTOR v1.0
1784304724.92,999,14,436,S,"999,14,436,S"
1784304725.00,1051,18,520,S,"1051,18,520,S"

Rules: META lines are filtered from all scripts if a file holds several runs (01 --append), the scripts keep only the run after the last reset header (# ACOUSTIC BIOREACTOR, or # BIOREATTORE ACUSTICO in older files). Record one file per block (Ch. 20).

12.3 Commands (PC -> Arduino)

The program accepts a single ASCII characters with no terminator required: S (sweep), B (burst), N (noise), R (raw), X (TX off) (see the limitation in Ch. 11.2). Unknown characters are ignored.

13. Software

13.1 Environment set-up

We reccomend Python >= 3.10.

Recommended virtual environment:

```
python -m venv .venv
# Windows: .venv\Scripts\activate   macOS/Linux: source .venv/bin/activate
```

requirements.txt: pyserial, numpy, pandas, matplotlib, scipy, opencv-python.

13.2 Acquisition - 01_data_from_sensor.py

The following functions are present within the program:

- **find_port():** Locates the port by searching for Arduino, CH340, USB Serial, ttyUSB.
- **start_logging(port, output_path, duration_s=None, split_s=None, append=False)**: Opens the port, waits 2 seconds for the microcontroller to reset, then sends the S command (the board starts idle). The reset header already in the serial buffer is kept and saved as META, so that appended runs can be separated (E-17). It refuses to write into an existing non-empty file

unless append is set (E-20). When it stops it sends X, so the transmitter goes silent (E-47). Writes the header, validates each line (four fields). Converts and adds pc_timestamp. Then saves. Flushes every 50 lines and displays a count every 10.

- **segment_path()** and **open_csv()**: Support the --split option, which rotates the acquisition to numbered files (session_001.csv, session_002.csv, ...) every N seconds. The header is written only once per file.
- **send_command(port, cmd)**: Sends a single command (S/B/N/R/X) and closes.

Argument	Example	Required	Function
--port	COM5	No (auto detected)	Find the correct port
--out	data/2026-09-20_anabaena_r1.csv	No (default data/session.csv)	Destination CSV file; the folder is created if missing.
--duration	900	No (default: unlimited)	Duration in seconds; automatic stop afterwards. Without it, records until Ctrl+C.
--split	300	No	Rotates into numbered files (..._001.csv, ..._002.csv) every N seconds.
--cmd	S	No	Sends a single command (S/B/N/R/X) and exits, without logging.
--append		No	Allows writing into an existing file (default: refuse). The analysis keeps only the last run of the file.

For light/dark protocols, do not use a fixed --split that is not aligned with the light timer. Either record one file per block, started at the transition and named after it (e.g. --out data/2026-10-01_r1_light1.csv --duration 900), or record one file per session and cut the blocks by pc_timestamp using a logged list of transition times.

Example:

python 01_data_from_sensor.py --port COM5 --out data/2026-09-20_anabaena_r1.csv --duration 900

13.3 Offline analysis: 02_analysis.py

The following functions are present within the program:

- **load_and_segment()** (acoustic_common.py): loads the CSV, keeps only the run after the last firmware header, removes META rows and incomplete sweeps, and returns one table per complete sweep.
- **normalize_amplitude()**: scales each sweep for display (minimum to 0, maximum to 1, relative dB floored at −40 dB).
- The file 02 computes no features: every feature comes from acoustic_common.py through 03 (Ch. 13.4). The plots cover 1000–3500 Hz.

Plot functions: plot_single_sweep, plot_waterfall, plot_baseline_comparison, print_statistics.
Arguments: --csv, --baseline <file.csv>, --show {all,sweep,waterfall,baseline}, --save.

Argument	Example	Required	Function
--csv	data/session.csv	Yes	Analysis of the CSV file produced by 01_data_from_sensor.py
--baseline	data/baseline.csv	No	Separate water-only CSV used as reference; needed only for --show baseline.
--show		No (default all)	Which plot to display: sweep, waterfall, baseline, all.
--save		No	Save the plots as PNG instead of opening windows.

The --show argument accept:

Value	Function
sweep	Show frequency response of the last sweep (raw + smoothed curve, peak and centroid marked).
waterfall	Show frequency x time colour map of the last 80 sweeps (log frequency axis).
baseline	Show a two-panel baseline <-> session comparison: mean baseline spectrum with +-1 SD band, and the difference of every session sweep from the baseline mean, coloured by time (red = first sweeps, green = last). Requires --baseline.
all	All applicable plots above.

The output files for --save are:

Plot	File name (from CSV stem)
Last sweep	plot_<stem>_last_sweep.png
Waterfall	plot_<stem>_waterfall.png
Baseline comparison	plot_<stem>_baseline_comparison.png

Example: python 02_analysis.py --csv data/session.csv --baseline data/baseline.csv

13.4 Sweep features: 03_features.py

Computes the features of Ch. 14.1 for each sweep and writes them to CSV only when --out is given (one row per sweep). Time columns: t_sec (mean Arduino time of the sweep, relative to the start of the file) and sweep_n. The table also contains state (LIGHT, DARK or NONE) and used (False for the sweeps excluded as settling). There is no pc_timestamp column: record one file per block (Ch. 20).

The dashboard uses fixed axes (1000–3200 Hz) to prevent autoscaling from visually amplifying the noise of a flat baseline.

Light state and settling: The state of a block is read from the file name, for example light... -> LIGHT, dark... or night... -> DARK and Any other name -> NONE. They can be also set with --state. For LIGHT and DARK files the first 180 s (--settle) are excluded from the statistics: the light switches at the start of the file and the first minutes are a coupling and thermal transient.

QC flag: For every file 03 prints QC = WARNING when |r(peak_ampl_db, centroid_hz)| >= 0.6 on the used sweeps, N/A with fewer than 20 sweeps, OK otherwise (Ch. 14.6).

S2S and coupling gate: for every file 03 also computes the S2S (Ch. 14.2) and the coupling gate (Ch. 14.6) on the used sweeps, prints them on the GATE line of the terminal summary and of the

dashboard, and writes them into the summary CSV. With --settle 0 it reproduces the whole-file values of this manual.

Summary: With --summary, 03 writes or replaces one row per block file in a summary CSV and prints the bracketed LIGHT − DARK comparison of its features (Ch. 19.3). Only blocks with QC OK and gate PASS enter the comparison.

The following functions are present within the program:

- **load_and_segment():** loads the CSV, removes the META rows, and breaks the session into individual sweeps.It is present in acoustic_common.py.
- **extract_feature_single_sweep():** calculates the following features from a sweep: centroid_hz, centroid_raw_hz, peak_freq_hz, peak_ampl_db, rms_abs_db, high_freq_ratio. It is present in acoustic_common.py.
- **extract_feature_session():** applies the previous function to all sweeps and adds the _slow components (centred moving average over 10 sweeps) and centroid_sigma_hz (rolling standard deviation of the centroid over the same window).
- **block_metrics(), passive_noise_adc():** S2S, robust S2S, gate quantities and passive noise of the block. They are present in acoustic_common.py.
- **plot_feature_dashboard()** plots the dashboard of features over time, with fixed axes.
- **qc_coupling(), print_summary():** QC flag and numeric summary of the block (used sweeps, centroid, median rolling σ, RMS, peak, high-frequency ratio, trends).
- **update_summary()**, **print_light_dark_comparison()**: with --summary, one row per block file in the summary CSV and the bracketed LIGHT − DARK comparison (Ch. 19.3).

Arguments: --csv, --out <feat.csv>, --save, --state, --settle, --summary.

Argument	Example	Required	Function
--csv	data/session.csv	Yes	Session CSV to process produced by 01_data_from_sensor.py
--out	feat.csv	No	Saves the per-sweep feature table as CSV
--save		No (default: shows the window)	Writes the dashboard as PNG (<stem>_features.png) instead of opening a window
--state	light	No (default: from the file name)	Set the light state of the block: light, dark or none
--settle	180	No (default 180 s)	Seconds excluded at the start of LIGHT and DARK files
--summary	summary.csv	No	Writes or replaces one row per file in a summary CSV and prints the bracketed LIGHT – DARK comparison

13.5 Real-time monitoring: 04_realtime_monitor.py

This is an experimental file, dedicated to real-time monitoring.

Example: *python 04_realtime_monitor.py --port COM5*

There are two threads (one serial and one Matplotlib animation) with thread-safe buffers.

This file finalizes each sweep and displays three panels: the spectrum of the last sweep (with its centroid) and the centroid and RMS energy of the last 40 sweeps. It is used while mounting the transducers, to see that the signal is present and settling. It records nothing and does not show the peak amplitude: the GO/NO-GO decision is taken on 5-min warm-up files analysed with 03 (Ch. 17.3).

The DataBuffer class (thread-safe, BUFFER_SWEEP_N = 40 sweeps in memory) finalizes each sweep in _finalize_sweep; the features are computed with the same function as 03 (extract_feature_single_sweep).

13.6 Operational sequence

1. Mount and couple the transducers, then use 04_realtime_monitor.py only to check that the signal is present, then close it to free the port.
   python 04_realtime_monitor.py --port COM5
2. Warm up with 5-min files until the GO/NO-GO criteria are met (Ch. 17.3). if the set-up changed, run the interference check (Ch. 18.5). Then record one file per block:
   python 01_data_from_sensor.py --port COM5 --out data/2026-10-01_elodea_r1/light1.csv --duration 900
3. After the session, analyse each block file (Ch. 19):
   python 03_features.py --csv data/2026-10-01_elodea_r1/light1.csv --out data/2026-10-01_elodea_r1/light1_feat.csv
   python 02_analysis.py --csv data/2026-10-01_elodea_r1/light1.csv --save

14. Metrics and algorithms

All features are calculated on the 60-point A(f) curve of each sweep, restricted to the 1000-3200 Hz band (≈ 55 of the 60 steps).

14.1 Sweep Feature

These are the columns produced by 03_features.py (one line per sweep). Each feature is a scalar descriptor that summarizes a property of the spectral response curve (an amplitude as a function of frequency) measured at each sweep, so that its temporal evolution can be followed without treating the 60 points of the spectrum individually. With each sweep, the system excites 60 increasing frequencies and measures their transmitted amplitude, obtaining a curve with maxima (peaks) corresponding to the resonances of the loaded system (Ch. 4.4). The features of this curve describe the location of the energy, its overall intensity, and its shape.

- **centroid_hz - spectral energy position:** This is the frequency center of the curve, i.e. the average of the frequencies weighted by the amplitude. A low centroid indicates energy concentrated in the low tones, a high centroid indicates energy shifted toward the highs. The transit of bubbles in the beam disrupts this position. However, the absolute value is dominated by coupling (replacing the transducers shifts it up to ~281 Hz) and therefore cannot be

interpreted in an absolute sense: only the intra-session variation is used. The S2S is not computed from this column but from the raw-amplitude centroid (Ch. 14.2).

- **peak_freq_hz - position of the dominant resonance**: This is the frequency of the curve's maximum, typically ~1550 Hz. It has diagnostic value: a marked shift generally indicates a change in the liquid level or geometry, not gaseous activity. It is not an activity metric; it is useful for understanding any systematic errors.
- **peak_ampl_db - coupling level:** $20 \cdot \log_{10}$ of the highest raw ADC amplitude in the band (≈ 43.5 dB at 150 ADC; the 110 ADC threshold of the gate is 40.8 dB). It describes the strength of the acoustic contact, not gaseous activity, and is comparable only within a session. It is the peak used by the QC flag of 03 (Ch. 14.6).
- **rms_abs_db - total signal energy:** It measures the energy of the entire curve, calculated on the raw (unnormalized) ADC amplitude. Bubbles attenuate the transmitted sound, reducing this value (−3.81 dB in the sparkling water test). Two conditions limit its use: it increases autonomously during warm-up due to mechanical settling (record only after the plateau) and, being an absolute quantity, it is comparable only within the same session.
- **high_freq_ratio - frequency energy distribution** It is the fraction of the normalized amplitude above 2000 Hz relative to the total (a sum of amplitudes, not of energy), ranging between 0 and 1. High values indicate energy shifted toward the secondary resonance cluster. Low values indicate energy concentrated on the main resonance. It is an exploratory shape descriptor: the granulometric interpretation (high <-> small bubbles, low <-> large bubbles) is a hypothesis not yet validated and further experiments are needed.
- **_slow components and centroid_sigma_hz - separation of time scales:** The file 03 adds two derived quantities.
  - The slow component is a moving average of 10 sweeps (≈ 1.7 min) and represents the slow trend. A slow change is attributed to biology only if it is absent in the medium-only blank
- centroid_sigma_hz is the standard deviation of the centroid over the same 10-sweep window. It rises with bubble-driven flicker, but also with any centroid change slower than one sweep and faster than ≈ 100 s, such as residual coupling motion or a lamp step It is the activity indicator of the dashboard and one of the features compared by --summary after s2s_hz (Ch. 19.3); it is not the S2S (Ch. 14.3).
- The first describes the direction of the process, the second its short-term variability.

In short, the centroid identifies the location of the spectral energy, the RMS quantifies its overall intensity, and the high-frequency ratio describes its frequency distribution. The slow components and centroid_sigma_hz separate the trend from the fluctuation.

Per-sweep features (extract_feature_single_sweep, band 1000–3200 Hz):

Column	Definition	Use
centroid_hz	Spectral centroid Σ(f·A)/Σ(A) on normalised amplitude. Only bins above MIN_AMPL_DB = −30 dB contribute.	Basis of the dashboard and of centroid_sigma_hz. Absolute value is coupling-dominated (drift up to 281 Hz between mounts). Remember to never read in absolute terms. Not used for the S2S, which is computed on the raw-amplitude centroid (Ch. 14.2).
peak_freq_hz	Frequency of the maximum of the normalised, smoothed spectrum: argmax_f A(f).	Diagnostic of the loaded resonance position (usually ~1550 Hz). This parameter shifts with liquid level/geometry. It is not an activity metric.

Column	Definition	Use
peak_ampl_db	$20 \cdot \log_{10}$(max A_raw): highest raw ADC amplitude in band, in dB (150 ADC ≈ 43.5 dB; 110 ADC = 40.8 dB).	Coupling level, comparable only within a session. Input of the QC flag of 03 (Ch. 14.6).
rms_abs_db	Absolute energy $10 \cdot \log_{10}$(mean A^2) on raw ADC, in band.	down with bubbles (attenuation, −3.81 dB with sparkling); up during warm-up. It is valid intra-session only; record after plateau.
high_freq_ratio	High-frequency amplitude fraction Σ_{f>2000} A / Σ A (HIGH_FREQ_THRESH = 2000 Hz).	Shape descriptor, exploratory. The grain-size reading (small/large bubbles) is an uncalibrated hypothesis, further experiments are needed.

Derived time-scale components (extract_feature_session, applied to centroid_hz, rms_abs_db, high_freq_ratio, peak_ampl_db):

Column	Definition	Use
*_slow	Centred moving average over SLOW_WINDOW_N = 10 sweeps.	Isolates the slow trend (biological signal). At ~10 s/sweep, 10 sweeps ≈ 1.7 min.
centroid_sigma_hz	Rolling standard deviation of centroid_hz over the same 10 sweeps (centred, at least 3).	Activity indicator of the dashboard and of --summary. Also responds to drift and steps slower than one sweep, not the S2S (Ch. 14.3).

This is the three-step sequence by which the spectrum curve is normalized. After these three steps, each spectrum goes from 0 to 1 (i.e. with 0 dB at the top). This erases information about how strong the signal was, but makes the shapes of different spectra comparable: two curves with the same shape but different heights become superimposable. This is useful for seeing where the peaks are, but not how high they are.

- Bring the minimum to zero: A <- A − min(A)
- Bring the maximum to one: A <- A / max(A)
- Convert to decibels: A_dB = $20 \cdot \log_{10}(A + \varepsilon)$.

Normalization removes the absolute level. Quantities that must keep it are computed on the raw ADC amplitude, before normalization: rms_abs_db, peak_ampl_db and the centroid used for the S2S (Ch. 14.2).

14.2 Primary metric: sweep-to-sweep variation of centroid (S2S)

The primary metric of the tool quantifies the instability of the spectral centroid between consecutive sweeps. Given a block of N sweeps under constant conditions, with centroids c_1 … c_N, the differences between adjacent sweeps are calculated $d_i = c_{i+1} - c_i$ and their standard deviation is taken as an index of activity:

$$S2S = std(d_1 \ldots d_{N-1})[Hz]$$

Centroid used for the S2S: The centroid of each sweep is weighted by the raw in-band ADC amplitude: c = Σ f·A_raw / Σ A_raw for 1000 <= f <= 3200 Hz, with no normalization and no amplitude threshold. All the reference values of this chapter were obtained with this definition. On the same blocks the centroid_hz column of 03 (normalised amplitude, bins above −30 dB) gives S2S values 2–4 times higher, because it amplifies the bins close to the noise floor, and in Trial 5 it ranks dark3 (25 Hz) above light1 (22 Hz). Do not compute the S2S from centroid_hz (E-46).

Slow coupling drifts produce small, nearly constant, consecutive differences that contribute negligibly to the standard deviation, while bubble transits in the beam generate centroid excursions that fall within one or two sweeps and dominate the S2S. The metric thus acts as a high-pass filter on the centroid, isolating the gaseous activity from the drift.

Validity condition: the S2S is not absolutely interpretable: a drop in peak amplitude during the block increases the burden of quantization noise on the centroid and inflates its standard deviation in the absence of bubbles (Ch. 14.5). Therefore, before each interpretation, a quality gate is applied to the coupling, which rejects the block if the correlation between peak amplitude (raw ADC) and the raw-amplitude centroid drops below −0.85. This indicates that the centroid is following the drift of the coupling rather than the bubbles (Ch. 14.3). The block is also rejected if the peak amplitude drops by more than 15 % during the block or if its mean is below 110 ADC (full criteria: Ch. 14.6). A block that fails the gate is classified neither as positive nor as null.

Implementation: in the software the S2S and the gate are computed by block_metrics() in acoustic_common.py, which 03 calls for every block (Ch. 13.4). The standalone listing below does the same on one block file (one file per block, Ch. 20). It reuses load_and_segment(), so the sweeps are exactly those read by 03, computes the raw-amplitude centroid and the peak of every sweep, and returns the S2S and the three gate quantities. The peak change compares the mean of the first three and of the last three sweeps, so that a single noisy sweep cannot decide the gate. Run it from the python/ folder; it reproduces the values of the table below.

```
import numpy as np
from acoustic_common import load_and_segment, ANALYSIS_BAND

def sweep_raw(sweep):
    """Raw-amplitude centroid (Hz) and peak (ADC) of one sweep, 1000-3200 Hz."""
    f = sweep['freq_hz'].to_numpy(float)
    a = sweep['ampl_adc'].to_numpy(float)
    b = (f >= ANALYSIS_BAND[0]) & (f <= ANALYSIS_BAND[1])
    return np.dot(f[b], a[b]) / a[b].sum(), a[b].max()

def block_metrics(csv_path, settle_s=0):
    """S2S and coupling-gate quantities for one block file (one file = one block)."""
    sweeps = load_and_segment(csv_path)
    t0 = sweeps[0]['t_sec'].iloc[0]
    sweeps = [s for s in sweeps if s['t_sec'].iloc[0] >= t0 + settle_s]
    c, p = np.array([sweep_raw(s) for s in sweeps]).T
    s2s = np.std(np.diff(c), ddof=1)
    r = np.corrcoef(p, c)[0, 1]
    change = (p[-3:].mean() - p[:3].mean()) / p[:3].mean()
    qc_pass = (r >= -0.85) and (change >= -0.15) and (p.mean() >= 110)
    return dict(n_sweeps=len(c), s2s_hz=round(s2s, 1), peak_mean_adc=round(p.mean()),
                peak_change_pct=round(100 * change), r_peak_centroid=round(r, 2),
                qc_pass=bool(qc_pass))
```

Reference values [M]: They are computed with block_metrics() on whole block files (03 --settle 0). The floor (no bubbles) is 2.4–5.8 Hz on blocks with good coupling: still water, organism-free baselines and dark blocks. A bubble effect is claimed when the S2S exceeds twice the floor of the same session. This was reached with sparkling water (≈ 6–7×), with the straw and with the plant at ≈ 10.7 bubbles/min (Trial 5, light1, 14.7 Hz, exploratory, this block fails the gate, Ch. 23.2), but not at ≈ 4 bubbles/min (Trial 5, light2, 4.6 Hz).

Condition	S2S (Hz)
Still water; baseline without organisms; blocks in darkness	2.4–5.8 (floor; in 05_elodea_1: baseline 5.5,, dark1 4.8, dark2 4.2, dark3 5.3)
Plant in light, ≈ 4 bubbles/min	≈ 4.6 (at floor: below LOD)

Condition	S2S (Hz)
Plant in light, ≈ 10.7 bubbles/min	≈ 14.7
Sparkling water	35–40 (≈ 6-7x the floor)
Straw	≈ 54 (6 sweeps only)

Property: The S2S responds to intermittent bubbles crossing the beam and is robust to slow coupling drifts. The S2S, however, is blind to continuous, steady-state bubbling (Ch. 14.4) and sensitive to signal-to-noise ratio confounding when the peak amplitude is low or decreasing (Ch. 14.5). These limits define the instrument's operating regime (Ch. 24) and are the reason for the QC gate controls.

14.3 Metric removed: block-wide centroid standard deviation

An alternative metric, used in early analyses, was the standard deviation of the centroid over the block (σ). It was removed because it was confounded by coupling drift: a slow drift that shifts the centroid by a few tens of Hz over the block produces a high σ even in the absence of bubbles. Unlike S2S, which operates on the differences between consecutive sweeps and attenuates slow variations (Ch. 14.2), σ integrates the entire centroid excursion, including the drift.

This was confirmed experimentally, with σ ≈ 28 Hz compared to only ≈ 4 bubbles/min measured on video, with a correlation between peak amplitude (ADC) and centroid equal to −0.97 within the block and −0.94 between blocks. A strongly negative correlation indicates that the peak and centroid move in a mirror-image fashion: the centroid is following the coupling drift, not the gaseous activity. This is the same criterion adopted by the QC gate (Ch. 14.2), which discards blocks with a correlation lower than −0.85.

The 6.3× factor reported for the sparkling water test was obtained with this metric. However, the result remains valid since the S2S on the same test provides ≈ ≈ 6–7× (34–40 Hz vs 5.8 Hz).

Rolling σ of 03 (centroid_sigma_hz): The file 03 still computes a standard deviation of the centroid, but over a 10-sweep (≈ 100 s) window. The short window removes most of the slow drift that invalidated the block-wide σ, but not changes on time scales from tens of seconds to minutes, such as residual coupling motion or a lamp step. It is therefore a screening indicator, not the primary metric. Across the trials the two measures disagree: centroid_sigma_hz is higher in light than in dark in Trial 6 (+5.0 Hz, 5 of 5 bracketed comparisons, p ≈ 0.013) and in Trial 8 (≈ 22 vs ≈ 12 Hz), where the S2S shows no light effect. In both trials the lamp raised the passive noise above 2 ADC (Ch. 23.4). A light effect seen only in centroid_sigma_hz is reported as unconfirmed.

14.4 Intrinsic limit: Blindness to continuous steady bubbling

The S2S measures fluctuations in the gas fraction in the beam, not its level. A continuous, uniform flow of fine bubbles shifts the centroid to a new mean value and keeps it stable there: the differences between consecutive sweeps become small again, and the S2S does not increase, even in the presence of gas. This results in a counterintuitive but expected behavior: a larger amount of gas in the continuous regime can produce a lower S2S than a smaller amount of gas in the intermittent regime.

The observation is experimentally documented [M]: in the face of excellent and stable coupling (peak amplitude 131–145 ADC) and a fine continuous stream of bubbles from the plant crossing the beam (Trial 8), the S2S in the light blocks remained at ≈ 4–5 Hz, at floor level. The QC gate does not detect

this condition (since the coupling is excellent) so the block would be formally interpretable and would be incorrectly classified as null. This is a limitation of the metric, not an artifact of the coupling.

Countermeasures:

1) Return the experiment to the validated regime. Reduce the light intensity so that gas production decreases and bubbles become sparse, larger, and intermittent, which is the regime in which S2S is validated (Ch. 14.2). The bubbling regime must be visually verified before recording (Ch. 18.2).
2) Adopt absolute metrics for continuous regimes. Evaluate the absolute centroid and attenuation (rms_abs_db) over short windows spanning sharp ON/OFF light transitions, with coupling certified as stable. The coupling must be as stable as possible.

14.5 SNR confounding

The centroid is an amplitude-weighted average of frequencies: its stability depends on the weights (the amplitudes of the 60 points) being well above the noise. As the peak amplitude drops, the entire spectrum approaches the ADC's quantization background (1 count in 1023) and so do the weakest bins: small random variations in amplitude translate into oscillations of the centroid from one sweep to the next. The S2S therefore increases in the absence of bubbles, due solely to the deterioration of the signal-to-noise ratio.

A block with a marked drop in peak amplitude is therefore not interpretable in any direction: a high S2S could be due to bubbles or noise, and the two causes are indistinguishable a posteriori. This is why the quality gate (Ch. 14.2) discards blocks with an average peak amplitude lower than 110 ADC or a drop of more than 15%, in addition to the peak-centroid correlation criterion.

14.6 Coupling quality gate

The quality gate is the criterion, applied during the analysis phase, that determines whether a block is interpretable. A block is excluded from interpretation if at least one of the following conditions is true:

Condition	Threshold	Confounding intercepted
Correlation between peak amplitude and raw-amplitude centroid within the block	$r < -0.85$	Coupling drift that drags the centroid (Ch. 14.3)
Decrease in peak amplitude between the start and end of the block	>15%	Signal-to-noise ratio degradation (Ch. 14.5)
Average peak amplitude in the block	<110 ADC	Insufficient signal-to-noise ratio (Ch. 14.5)
Number of sweeps in the block	< 20	Correlation and S2S not meaningful
Passive noise in the block (median std of P rows per pause, TX off)	>2 ADC	Electrical interference (lamp) or mechanical agitation (Ch. 18.5)

The first three conditions are complementary: the first intercepts the centroid that follows the coupling rather than the bubbles, while the other two intercept the low signal-to-noise regimes in which the S2S is inflated by quantization noise. The peak amplitude is the highest raw ADC amplitude in the band and file 03 reports the same quantity in dB (peak_ampl_db, 110 ADC = 40.8 dB).

The gate is pre-registered (the passive-noise threshold is still provisional). The thresholds are set before examining the data and are applied to all blocks without exception before calculating or

interpreting the S2S. This prevents the a posteriori selection of favorable blocks and makes the classification reproducible. The thresholds were derived from Trial 5 and are pre-registered for every later session. Applied retroactively they exclude the Trial 5 light1 block (mean peak 68 ADC, Ch. 23.2), which is why Trial 5 is reported as exploratory.

But be careful. The gate guarantees that a block is acoustically reliable, not that a null outcome is biologically null. A block can pass the gate and remain uninformative if the bubbling is continuous and stationary (Ch. 14.4).

Gate and QC flag of 03: File 03 prints its own QC flag per file: WARNING when |r(peak_ampl_db, centroid_hz)| >= 0.6 on at least 20 used sweeps, N/A with fewer sweeps (Ch. 13.4). The flag is a screening rule set heuristically on Trials 3–8. It is two-sided, uses the normalised centroid and does not test the peak level, the peak change or the passive noise. 03 prints both for every block (QC and GATE lines) and uses a block in the light–dark comparison only if it passes the gate of this section and its QC flag is OK. Until a single rule is pre-registered, both are required.

Positive controls with dense bubbling: Gas attenuates the transmitted signal (sparkling water: peak −40 %), so the peak criteria also respond to the bubbles themselves. For abiotic positive controls the coupling is verified on the bubble-free blocks recorded before and after (Ch. 18.1, 22).

14.7 Experimental features removed

The following functions are older versions of the pipeline that calculated four additional features. These features were later removed because they were exploratory and never demonstrated a correlation to a physical or biological reference:

- **rms_db**: This is the RMS calculated on the normalized spectrum (shape energy). It is redundant with rms_abs_db, which measures the absolute energy on the raw ADC (Ch. 14.1) and is the only useful version for attenuation. The normalized version does not add information.
- **bandwidth_hz**: This is the spectral bandwidth. It is a shape descriptor without a validated interpretative threshold.
- **spectral_flatness**: This measures how "flat" (noisy) the spectrum is and how peaked (tonal) it is. This is an exploratory feature and has never been correlated to a physical quantity of the system.
- **kurtosis_spectral**: This is the kurtosis of the spectral distribution. Like the previous feature, it was conceived as an experimental function, but has never been verified.

The calculation logic is preserved and commented in the code, so that it can be quickly reactivated if validated in the future. Including it at present would produce numbers without a physical or biological basis.

PART III: USE

15. Safety

15.1 Electrical

The system operates at 5 V safety extra-low voltage (SELV), powered by the computer's USB port. There is no electrical hazard for the operator. The TX transducer is driven through a 100 Ω limiting resistor, which limits the current delivered to the capacitive load.

Precautions:

- Do not power the board with voltages exceeding the nominal voltage.
- Do not connect the transducers to external signal generators without current limitation.
- Connect and disconnect the transducers and wiring with the board unpowered (USB disconnected) to avoid transients on the analog input.
- Use only the computer's USB power supply or a compliant power supply. Do not modify the front end while it is powered up.

Note that mains-powered auxiliary equipment (like lamp, timer, orbital shaker) is not SELV: keep it away from liquids and connect it to an outlet separate from the computer (if possible). This also reduces the interference coupled into the RX input (Ch. 18.5). Run the laptop on battery during recordings: a charger connected to the mains closed a ground loop through the USB cable (Ch. 23.7).

15.2 Chemical

The coupling and cleaning agents must be compatible with the transducers (brass, tin solder, silver electrode), the flask (polystyrene) and the holder (PLA or PETG). We can use:

- Sodium hypochlorite (bleach): inside of the flask only, followed by thorough rinsing. Never on piezos, solder joints or pins (it corrodes brass, tin and silver).
- Ethanol / isopropanol: piezos, cables and electronics.
- Polystyrene flask wall (outside): 70 % ethanol only, without rubbing. Concentrations >= 90 % craze and crack the polystyrene under clamp load.
- Hydrogen peroxide: preferred for the inside of the flask (less toxic residues for the culture). Avoid contact with metals.
- Holder: PLA tolerates alcohol but not immersion in bleach. Is better to use PETG that is more tolerant.
- Any solvent removes the couplant: after cleaning, redo coupling, warm-up and the GO/NO-GO gate.

15.3 Biological

Cyanobacterial cultures: handle according to the laboratory biosafety rules and inactivate before disposal (autoclave or hypochlorite inside the flask). Elodea canadensis is non-native in Europe: never release plant material or water from the experiments into the environment; dispose of them as waste.

15.4 Optical

The 100-300 µmol m^{-2} s^{-1} illumination LEDs are intense at the operating distance so avoid prolonged direct eye exposure.

16. Cleaning, maintenance and chemical compatibility

16.1 Principle

Remember that the transducers are external and never come into contact with the culture. Cleaning and sterilization operations therefore involve two different parts: the inside of the flask (biology) and the sensor surfaces (outer wall, transducers, support, wiring). Each part therefore requires different reagents.

16.2 Materials and reagents

Component	**Ethanol/isopropanol**	**Sodium hypochlorite**	**Hydrogen peroxide (3–10%)**
Interior flask (sterilization)	Suitable (70% ethanol; evaporate)	Suitable with abundant rinsing (toxic residues for the culture)	Suitable, preferred (decomposes into H_2O and O_2, release less toxic residues). Remember to rinse
Outer wall T25 (polystyrene)	70% ethanol without rubbing. Concentratuion >= 90% causes clouding and crazing under load	Suitable with rinsing and drying	Suitable
Transducers: brass, solder, metallization	Suitable (evaporates without residue)	Unsuitable: Chlorinated oxidizer, corrodes brass, tin, and silver	with caution because cause oxidating, after the treatment rinse immediately
Wiring (PVC), connectors	Suitable but dry the contacts	Unsuitable because corrodes the pins	with caution (leaves moisture)
Electronics (Arduino, LM358)	Suitable	Unsuitable	Unsuitable (moisture, residues)
PLA support	Suitable	Avoid immersion	Suitable
PETG support	Suitable	Suitable for short passes	Suitable
Coupling	Removed by any solvent, after you need to repeat coupling, warm-up, and GO/NO-GO	Removed by any solvent, after you need to repeat coupling, warm-up, and GO/NO-GO	Removed by any solvent, after you need to repeat coupling, warm-up, and GO/NO-GO

16.3 Inter-session procedure

A standard procedure is the following:

1. End session: X, then stop the logger and save the notes.

2. Loosen the screws and remove the flask.

3. Remove the couplant from the wall with paper, then wipe with 70% ethanol or isopropanol. From the transducers, only use paper and, if necessary, isopropanol.

4. Sterilize the inside of the flask with hydrogen peroxide or hypochlorite, followed by a thorough rinse with sterile water. Remember to dry the outside wall thoroughly before reassembly.

5. Visually inspect the transducer solder joints and gently pull the wires.

16.4 Periodic maintenance

Recommended maintenance (at the discretion of the user):

Interval	Operation
Every session	Check the plateau peak and note the couplant used
Every 5 sessions	Check the bias (2.5 V on pin 3) and the continuity of the transducer wires
Every transducer replacement	Repeat the abiotic acceptance test (Ch. 18.1)

17. Standard Operating Procedure

17.1 Preparation checklist

This is a small checklist where you can find the main points to keep in mind for the preparation of the experimental apparatus.

- Flask with clean and dry outer walls and liquid volume measured and noted.
- Transducers in holder with wires intact.
- Couplant in a thin film on the wall and disk: vaseline only for sessions shorter than ≈ 20 min after warm-up; for longer protocols silicone grease or ultrasound gel (Ch. 10.3).
- Flask lip-sealed with contact screws.
- Liquid temperature noted (<= 25°C for biological experiments).
- Illumination: LED fixed with distance and intensity noted.
- Darkening available (no stray light).
- Laptop on battery (charger unplugged) or galvanic USB isolator; mixer off.

17.2 Experiment

The verification and execution process is divided into several phases. A session is everything recorded without moving the transducers; all comparisons are made inside the same session. It runs in this order:

1. Mount and couple the transducers (Ch. 9, 10). Check with 04_realtime_monitor.py that the signal is present, then close it (Ch. 13.5).

2. Warm up and take the GO/NO-GO decision (Ch. 17.3).

3. If the lamp, the mixer, the bench or the wiring changed, run the interference check (Ch. 18.5).

4. Record the blocks of the protocol (Ch. 18): one file per block, started at the change of condition and named after it (Ch. 20). Do not touch the set-up between blocks.

5. During each block, apply the rules of Ch. 17.4 and log every event with its time.

6. End the session (Ch. 16.3) and analyse the block files (Ch. 19).

It is recommended to consult the "User Guide - Acoustic Sensor" file for further details.

17.3 Warm-up and GO/NO-GO

After clamping, the couplant settles under the pressure and the received signal drifts on its own for several minutes. Data recorded in this phase describe the contact, not the bubbles. Recording starts only when the peak amplitude has reached a plateau (GO).

1. Record 5-min files: *python 01_data_from_sensor.py --out data/<session>/warmup1.csv --duration 300*.

2. Analyse each file with *python 03_features.py --csv data/<session>/warmup1.csv* and apply the criteria below.

3. If a criterion fails, record warmup2.csv, warmup3.csv, ... and repeat. If the peak has not settled after ≈ 20 min, remove the transducers, clean them, re-apply the couplant and start again (NO-GO).

Criterion (latest warm-up file)	**GO**	**NO-GO**
Peak trend over the last 3–5 min	Flat: a few ADC (≈ +- 0.2 dB), no monotonic trend	Still rising or falling
Mean peak level	>= 110 ADC (>= 40.8 dB); typically 130–200 ADC	< 110 ADC, ≈ 10 ADC means no contact
QC flag of 03	OK	WARNING
P rows in the terminal of 01 (TX off)	Vary by +- 2-3 counts at most (std <= 2 ADC)	Larger fluctuations: find the source first (Ch. 18.5)

Short files are used because 03 cannot read a file that 01 is still writing, and because they document the warm-up. The live monitor (04) does not show the peak amplitude and is not used for this decision.

17.4 Recording

You can film the whole session to count, as a visual reference, the bubbles produced by the culture or the plant (in our case Elodea) (Ch. 18.4). During each block follow these rules:

- Do not touch the flask, the holder, the cables or the bench while a block is recording. Any contact ends the block. If this is the case, note the time and record the block again in a new file.
- Stop the session (NO-GO) if the peak falls below 110 ADC or keeps falling: the couplant is creeping (E-07).
- Log every event with its time: light switching, temperatures at the start and end of each block, video synchronisation, anomalies.
- Keep the mixer off. If the culture must be mixed, mix between blocks and wait 2–3 min before the next block.

18. Experimental protocols

Every protocol has the same structure: one session with a fixed mount, blocks of constant condition recorded as separate files, and comparisons made only inside the session. Durations are those of Trials 4–8: 15-min blocks of 90 sweeps, of which 72 are used after the 180-s settling exclusion.

18.1 Abiotic acceptance test

Purpose: show that the instrument responds to free gas in the beam, without biology. Run it on a new build, after every transducer replacement (Ch. 16.4) and before every biological campaign.

1. Fill the flask to the marked level with still bottled water, couple and warm up to GO (Ch. 17.3).

2. Record still1.csv (10 min).

3. Without moving the holder, empty the flask through its neck, refill to the same level with sparkling water from a freshly opened bottle and record sparkling1.csv (10 min). The water degasses with time so use the first 10 min.

4. Empty, refill with still water and record still2.csv (10 min). The still blocks before and after bracket the sparkling block and show whether the mount changed.

5. Optional coarse-bubble check: blow gently through a straw immersed in still water for ≈ 1 min (straw.csv).

Acceptance: S2S of the sparkling block >= 2 x the S2S of the still blocks (reference, Trial 4: 34–40 Hz against 5.8 Hz). The still blocks pass the coupling gate and the mean peak of still2 is within 15 % of still1. The sparkling block is expected to fail the peak criteria, because the gas attenuates the signal (Ch. 14.6).

18.2 Biological validation with Elodea canadensis

Purpose: detect light-driven O_2 bubbling from a living organism with video as an independent reference. Elodea is used because its bubbling is visible and controllable. Its bubbles are larger than those of a microbial culture (Ch. 4.1).

Preparation:

- Medium: dechlorinated water with $NaHCO_3$ (≈ 10 mM, the same amount in every session), so that light, not carbon, limits photosynthesis.
- Acclimate the shoot for ≈ 15 min under the lamp in the same medium, then re-cut the stem under water just before mounting and place it cut end up so that the bubbles leave from a defined point.
- Fill to the marked level, leave the cap resting and not closed so that the gas can vent.
- Align the bubble stream with the TX–RX beam (the liquid region between the discs, Ch. 4.4) and fix the camera (Ch. 18.4). Water temperature <= 25 °C, noted at the start and end of each block.

Bubbling regime (REQUISITE): Before recording, watch the shoot in light. The S2S is validated only for sparse, intermittent bubbles (Ch. 14.4). If a fine continuous stream forms, reduce the light intensity or increase the lamp distance until the bubbles are separate events, and note the setting.

Block design:

1. Optional no-plant baseline (medium only, 5 min), if the shoot can then be inserted without moving the holder.

2. Warm-up with the light on, not analysed, until GO (Ch. 17.3).

3. Alternate dark and light blocks of 15 min, starting and ending with dark (D–L–D–L–D–L–D: at least three light blocks bracketed by dark blocks, the file 03 also accepts the L–D–L design of the User Guide and uses whichever gives more bracketed comparisons), one file per block (dark1.csv, light1.csv, ...).

4. Video during every light block, with a synchronisation event at each file start (Ch. 18.4).

Controls: (i) Medium-only blank under the same light schedule, to exclude lamp artefacts; (ii) heat-killed shoot under the same schedule, to exclude thermal or photo-acoustic effects. A light effect is attributed to biology only if it is absent in both.

Pre-registered success criteria: This criteria are fixed before the session:

- Every analysed block passes the coupling gate and the QC flag (Ch. 14.6)
- In every bracketed comparison the light block has a higher S2S than the mean of its two neighbouring dark blocks
- At last one light block exceeds 2 x the dark floor
- Blocks with more bubbles on video have a higher S2S
- No comparable effect in the controls

18.3 Cyanobacteria Nostoc (Anabaena) sp. PCC 7120 in the T25 flask

The protocol is defined and was run in three sessions, none yet interpretable for biology. Each removed one instrumental problem: lamp-switching steps and mixer noise on 23/09 (Ch. 23.5), lamp kept on but the shaker still running on 05/10 (Ch. 23.6), and on 12/10 the shaker was off but electrical noise remained; a follow-up test traced it to the laptop charger (Ch. 23.7).

- Culture in BG-11 with ≈ 10 mM $NaHCO_3$, OD_{750} 1.0–1.5 at the start (measured and logged); illumination 150–300 µmol m^{-2} s^{-1}; temperature <= 25 °C.
- Interference check passed on the same bench, with the same lamp, before the session (Ch. 18.5). Laptop on battery (or a galvanic USB isolator) and mixer off during recordings. Mix between blocks only and then wait 2–3 min.
- Block design as in Ch. 18.2 (at least three light blocks bracketed by dark), plus a medium-only blank recorded with the same light schedule.
- Replicates: >= 3 technical replicates (light/dark cycles inside a session) and >= 3 biological replicates (independent cultures, separate days, full remount). Cycles are not biological replicates.
- The sensor sees only the gas phase (Ch. 3). A dissolved-oxygen probe, when available, confirms that the medium exceeds 100 % saturation. Without it a null result cannot distinguish “no production” from “production below supersaturation”.
- Microbial bubbles are sub-millimetre and resonate far above the band (Ch. 4.1): the expected signal is smaller than with Elodea.

18.4 Video reference

You can use this method to verify the presence of bubbles in the culture. The workflow is:

- Camera on a tripod, framing the region between the transducers. Put a black background behind the flask and side lighting. The camera must not move during the session.
- Synchronisation: at each file start make a visible event (a hand or an LED in frame) and log its time. The video is aligned to pc_timestamp through it.
- Count the bubbles per second with video-analysis software (we used an OpenCV script, one row per second as in bubbles_light1.csv)
- Inspect any minute with an outlying count with --debug before using it. In Trial 5 a burst of ≈ 130 counts in one minute was a ROI artefact (2 real bubbles) and was excluded.
- The photosynthesis lamp also lights the video: dark blocks cannot be counted without an infrared illuminator and an IR-capable camera. Without IR video, bubble counts in dark blocks are assumed, not measured.

18.5 Interference check

Requisite: Run an interference check before every biological session and whenever the lamp, the timer, the mixer, the bench or the wiring changes. Run the laptop on battery. On 23/09/2026 lamp switching produced steps in all features, blank included, and the mixer raised the noise floor to the level of a real bubble signal (Ch. 23.5). This are the step:

1. Flask with water or medium only, coupling at plateau, all equipment in its recording position.

2. Record 15-20 min in sweep mode:

python 01_data_from_sensor.py --out data/<session>/interference.csv --duration 1200.

3. Every 5 min switch the lamp off or on, then, separately, the mixer. log the times.

4. Open the dashboard of 03: at the switching times the RMS and peak panels must show no step.

5. Compute the passive noise: standard deviation of the P rows of each 5-s pause, median over the pauses of each state (Appendix C).

Acceptance: passive noise <= 2 ADC in every state, and no step in rms_abs_db, peak_ampl_db or in the bins above 3200 Hz at the transitions.

Mitigation: Run the computer on battery or through a galvanic USB isolator: on 13/10/2026 the laptop charger was the main noise path (Ch. 23.7). Put lamp and timer on a mains outlet separate from the computer. Keep the lamp cables away from the RX wire. Use a short, shielded RX cable (with a ferrite if needed), or run the computer on battery. Keep the mixer off during recordings.

19. Data analysis

The operating procedure for the data analysis is explained in detail in the “User Guide – Acoustic Sensor”. The following subchapters summarise the main points.

19.1 Per-block processing

1. Dashboard and feature table of each block:

python 03_features.py --csv data/<session>/light1.csv --out data/<session>/light1_feat.csv --summary data/<session>/summary.csv --save.

2. S2S and coupling gate of each block: GATE line of 03 (block_metrics(), Ch. 14.2 and 14.6).

3. Passive noise of each block (Ch. 18.5, Appendix C), also on the GATE line.

4. Spectra and waterfall, when needed:

python 02_analysis.py --csv data/<session>/light1.csv --save

19.2 Classification of the blocks

The gate is applied before any metric is read (Ch. 14.6). Each block then falls into exactly one class:

Class	Condition	Meaning
Not interpretable	Fails the gate, QC flag WARNING or N/A, or passive noise > 2 ADC	Reported, never used as positive or null

Class	Condition	Meaning
Positive	Passes; S2S >= 2 x the dark (or no-organism) floor of the same session	Intermittent gas activity detected in the beam
Null	Passes; S2S < 2 × that floor	No intermittent bubbling detected. Check the regime (Ch. 14.4) before calling it biologically null

19.3 Light–dark comparison

Light and dark blocks are compared in brackets: each block is compared with the mean of its two neighbours of the opposite state: a dark block with the light blocks before and after it (L–D–L), or a light block with the dark blocks before and after it (D–L–D, Ch. 18.2). 03 uses the design that gives more comparisons. A linear drift across the session cancels in this difference, whereas a plain comparison of a light block with the next dark block would read the drift as a light effect. At least three bracketed comparisons are required. Report their number, the mean difference, its standard deviation and how many differences are positive (e.g. 5/5). With --summary, 03 performs this comparison on its features, with s2s_hz first and only on blocks with QC OK and gate PASS: the p value it prints (one-sample t-test) is approximate because neighbouring comparisons share light blocks. Blocks from different sessions are never combined: 03 skips the comparison when the summary contains more than one folder.

20. Data management

Here are some tips for managing session results:

- One folder per session: data/YYYY-MM-DD_<organism>_r<n>/.
- One file per block, started at the change of condition and named after it: warmup1.csv, baseline.csv, blank.csv, light1.csv, dark1.csv (or night1.csv). 03 reads the light state from the beginning of the name. Do not use a fixed --split for light/dark protocols.
- Session log in the same folder (log.txt): date and operators, flask and fill volume, couplant and clamp setting, lamp, distance and intensity, organism and density, temperatures, code version, time of every event (block starts, light switching, synchronisation events, anomalies).
- Raw CSV files are never edited. Never open and save them in Excel: re-saving quotes the rows and makes the file unreadable by the scripts. Work on copies.
- Videos and bubble-count files in the same folder, named after their block (bubbles_light1.csv).
- Analysis outputs (feature tables, summary CSV, PNG) kept apart from the raw files and labelled with the git commit of the code used.
- Back up the session folder at the end of the day.

PART IV: CHARACTERIZATION AND VALIDATION

21. System characterization

21.1 Transmission spectrum and analysis band

The transmission spectrum of the water-filled T25 flask shows the resonances of the loaded system: a main doublet at ≈ 1550/1583 Hz and secondary peaks at ≈ 2120 Hz and 2454/2558/2612 Hz (Ch. 4.4). Above ≈ 3.2 kHz only the noise floor (≈ 10 ADC) remains, which sets the analysis band at 1000-3200 Hz (Ch. 11.4). The main resonance moves with the fill level and the mount: ≈ 1206-1367 Hz in Trial 5, against 1550/1583 Hz in the characterization. Absolute frequencies are therefore never compared between sessions.

21.2 Coupling

Without couplant the peak is ≈ 11 ADC, at the level of the background. With couplant it reaches ≈ 150 ADC (Ch. 10.1). After clamping, the peak drifts for 10–20 min (warm-up). With vaseline it keeps decreasing afterwards: −42 % in ≈ 28 min in a dedicated test and −35 % from the baseline block in Trial 6 (E-07). The coupling state is the main source of non-biological variation of the centroid (Ch. 14.3, 14.5).

21.3 Repeatability and reproducibility

Quantity	Value [M]	Condition
Centroid, within a run	σ ≈ 2–4 Hz	Still water, fixed mount
Resonance and centroid between mounts	up to ≈ 281 Hz	Transducers removed and re-mounted
S2S floor	2.4–2.8 Hz	Still water, Trial 3 (3 blocks, peak 159–174 ADC)
S2S floor	3.6–5.8 Hz	Still water, no-plant and dark blocks that pass the gate, Trials 4, 5, 7, 8
S2S with degraded coupling	6.4–7.9 Hz	Dark blocks of Trial 6 (peak 68–105 ADC)

Intra-session comparisons are meaningful at the level of a few hertz, values from different mounts are not comparable. The holder (Ch. 9) is intended to reduce the remount spread, its test is in progress.

21.4 Timing

One sweep every 10.02 s (median interval between SWEEP_START markers): ≈ 5 s of sweep, then 5 s of passive listening. A 15-min block contains 90 sweeps, 72 remain after the 180s settling exclusion of 03. Bubbles that cross the beam only during the passive pause are not seen by the sweep.

22. Abiotic validation

The trial 4 (22/07/2026) compared still and sparkling water in the same session, with the transducers untouched. S2S computed as in Ch. 14.2:

Block	Sweeps	S2S (Hz)	Mean peak (ADC)
Still water (baseline_for_sparkling)	60	5.8	152

Block	Sweeps	S2S (Hz)	Mean peak (ADC)
Sparkling water (sparkling)	60	34.3	94
Sparkling water, ≈ 10 min later (sparkling_pt2)	60	40.4	88
Straw (07_elodea_3/straw.csv, Trial 7)	6	54.2	157

Sparkling water raises the S2S 6–7 times above the still-water floor of the same session, attenuates the received energy (rms_abs_db −3.81 dB) and the peak (−40 %), and shifts the mean centroid by ≈ +80 Hz [M]. The sparkling blocks fail the peak criteria of the gate by construction, because the gas itself attenuates the signal (Ch. 14.6); the coupling was checked on the still-water block. The straw run is too short (6 sweeps) for statistics and serves only as a coarse-bubble check.

ERROR (lesson learned): A first comparison between different sessions gave a centroid shift of −201 Hz. The same-session control showed that it was mostly an inter-session artefact (Ch. 21.3). Since then every comparison is made inside one session.

23. Biological validation

23.1 Overview

Trial	Date	Design	Peak (ADC)	Video	S2S light / dark (Hz)	Outcome
5 05_elodea_1	17/08/2026	No plant baseline, warm-up, D–L–D–L–D	68–210	light1 10.7, light2 4.0 bubbles/min	14.7 and 4.6 / 4.2–5.3	Exploratory positive (light1), light1 fails the gate
6 06_elodea_2	19/08/2026	Baseline, warm-up, 6 × (L–D)	112 → 68	≈ 20–30 bubbles/min	5.5–7.5 / 6.4–7.9	Not interpretable: coupling decay
7 07_elodea_3	20/08/2026	Warm-up, straw, 4 × (L–D)	159–196	None	4.5–5.6 / 3.6–5.1	Null, production judged insufficient
8 08_elodea_4	22/08/2026	Warm-up, 2 × (L–D), ON/OFF/ON file	131–145	None	4.0–4.4 / 4.7–5.3	Null in the blind regime (continuous bubbling)

23.2 Trial 5 (05_elodea_1)

This was one shoot, one session, with fixed mount and video during the light blocks. Metrics from block_metrics() on whole block files:

Block	Bubbles/min (video)	S2S (Hz)	Mean peak (ADC)	Peak change	r(peak, centroid)	Gate
baseline (no plant)	0	5.5	170	0 %	−0.29	pass
dark1	not filmed	4.8	79	−19 %	−0.82	fail (level, change)
light1	10.7	14.7	68	−13 %	−0.67	fail (level)
dark2	not filmed	4.2	73	+7 %	−0.21	fail (level)
light2	4.0	4.6	124	−37 %	−0.97	fail (r, change)
dark3	not filmed	5.3	210	−13 %	+0.67	pass

What it shows: In light1, with ≈ 10.7 bubbles/min on video, the S2S rose to 14.7 Hz, about three times the level of the dark and no-plant blocks (4.2–5.5 Hz), and returned to the floor in the following dark block. At ≈ 4 bubbles/min (light2) it stayed at the floor. Within light1, acoustic variability and bubbles per minute correlate moderately (Pearson 0.59, Spearman 0.35). The first bubbles appeared ≈ 30 s after the light was switched on.

Limits: One shoot and one session: the cycles are technical, not biological, replicates. Only one of the two light blocks is positive. The block-level correlation with the video (r = 0.91 over six blocks) is driven by light1 with zero buble in the dark block. The S2S metric and the gate were chosen after these data had been seen, and the gate excludes light1 (mean peak 68 ADC < 110). The S2S of light1 is also driven by a few sweeps: removing the single disturbed sweep at ≈ 510 s lowers it from 14.7 to 10.2 Hz, and its robust version (1.4826 x MAD of the sweep-to-sweep differences, printed by 03) is 5.5 Hz, at the level of the dark blocks (4.7–5.9 Hz). The video counts 5 bubbles in the window of the two largest jumps. Trial 5 is therefore a proof of principle with exploratory status, not a confirmatory validation.

23.3 Trials 6–8: characterized failure modes

Trial 6, coupling decay: Vaseline cold-flowed for the whole session: the mean peak fell from 112 ADC (light1) to 68 ADC (last dark block), −35 % from the baseline block. Eleven of the twelve light and dark blocks fail the gate, and the S2S shows no light effect (5.5–7.5 Hz in light, 6.4–7.9 Hz in dark), although the video shows ≈ 20–30 bubbles/min in every light block. centroid_sigma_hz is higher in light in 5 of 5 bracketed comparisons (+5.0 Hz, p ≈ 0.013), but the lamp raised the passive noise from 1.7 to 4.2 ADC in the light blocks (Ch. 23.4), so this difference cannot be attributed to the bubbles. The session was not interpretable. It shows that the warm-up and coupling checks are mandatory.

Trial 7, no detectable activity: Coupling excellent and (159–196 ADC, every light and dark block passes the coupling criteria of the gate, but the light blocks exceed the provisional 2-ADC passive-noise criterion, Ch. 23.4). S2S 4.5–5.6 Hz in light and 3.6–5.1 Hz in dark, centroid_sigma_hz is lower in light in all three bracketed comparisons. Bubble production was judged insufficient. The outcome was null, with the bubbling regime not documented.

Trial 8, blind regime: Coupling excellent (131–145 ADC, every light and dark block passes the coupling criteria of the gate, but the light blocks exceed the provisional 2-ADC passive-noise criterion, Ch. 23.4) and a fine continuous stream of bubbles crossed the beam in light. The S2S stayed at the floor (4.0–4.4 Hz in light, 4.7–5.3 Hz in dark), as expected for steady bubbling (Ch. 14.4). centroid_sigma_hz was higher in light (≈ 22 vs ≈ 12 Hz, two cycles), while the lamp raised the passive noise from 1.7 to 3.8 ADC. The final ON/OFF/ON file (alternate_4) is not interpretable: its peak fell by 30 % (r = −0.90). Outcome: the S2S is blind to continuous bubbling; the absolute metrics of Ch. 14.4 need a dedicated test.

23.4 Lamp interference in the Elodea trials

Trial	Passive noise, dark (ADC)	Passive noise, light (ADC)	Ratio
5	1.3	1.8	× 1.4
6	1.7	4.2	× 2.4
7	1.7	2.9	× 1.7
8	1.7	3.8	× 2.2

Median over the pauses of each block of the standard deviation of the P rows (TX off). The lamp couples electrical noise into the receiver in every trial. In Trials 6–8 the light blocks exceed the 2-ADC acceptance level of the interference check (Ch. 18.5). The swept amplitudes are affected too, so a light-minus-dark difference measured in these trials can contain a lamp contribution. No blank or heat-killed control was recorded under the same light schedule, both are required in the confirmatory session (Ch. 18.2).

23.5 First cyanobacteria session (23/09/2026)

A medium-only blank and two independent cultures were recorded in separate mounts, with the light on an automatic timer and an orbital mixer running for the whole day. Coupling was adequate (130–160 ADC). The session is inconclusive for biology because of two instrumental effects. (i) Lamp switching produced square-wave steps in all features of 03, larger in the blank than in the cultures (rms_abs_db ON − OFF: +6.6 dB in the blank, +3.2 and +1.4 dB in the cultures), with a uniform pedestal also above 3200 Hz. (ii) The mixer raised the passive noise to 4–11 ADC with the lamp off and the rolling σ of the centroid to 21–35 Hz at rest, a level comparable to sparkling water. In addition, the 15-min --split files were not aligned with the light timer (E-44). The corrective actions are in Ch. 18.3, 18.5 and 20.

23.6 Second cyanobacteria session (05/10/2026)

We remove the lamp-switching steps seen on 23/09 (Ch. 23.5). The lamp stayed on for the whole session, and light and dark blocks were made by opening and closing an opaque box around the flask.

This session has medium-only blank (1 dark + 1 light block) and two cultures of Nostoc (Anabaena) sp. PCC 7120 in separate mounts: Sample 1 (OD_{750} 0.445, D–L–D–L–D–L) and Sample 2 (OD_{750} 1.140, D–L–D–L). Blocks of 15 min (90 sweeps), one file per block. The shaker was running during the recordings (its speed was raised by mistake during Sample 1, light 1), and the Sample 1 flask was shaken by hand between the first blocks.

Every block exceeded the 2-ADC passive-noise criterion (5–12 ADC). By our pre-registered rules the session is not interpretable for biology; the observations below are descriptive.

What the session shows:

- Keeping the lamp on removes the switching steps**;** we see that in the blank, light − dark RMS was −0.46 dB, against +6.6 dB on 23/09.
- Opening and closing the box raised, in the blank, the passive noise from 5.3 to 7.2 ADC and the centroid by ≈ 15 Hz: any light − dark difference contains this effect.
- With the shaker running, the S2S floor was 16–35 Hz: the Elodea positive block (14.7 Hz) would have been invisible.
- Sample 1 had a lower S2S in light in 3/3 bracketed comparisons (−7.1, −3.2, −1.7 Hz). Sample 2 had a higher one (+3.0, +2.4 Hz), with the noise and the centroid rising as in the blank. centroid_sigma_hz showed no difference.

Corrective actions (applied on 12/10, Ch. 23.7): shaker off during recordings and on another bench. Then we mande an interference check before the culture, a darkening method that does not change the acoustic or electrical environment, checked in the blank over >= 2 cycles. Sequences ending with dark (D–L–D–L–D–L–D) to obtain three bracketed comparisons. We then create block files named light1.csv, dark1.csv in one folder per sample (or --state). The acoustic_commons.py was updated so both headers were accepted.

23.7 Third cyanobacteria session (12/10/2026)

We repeat the cyanobacteria test with the corrections from the previous sessions (Ch. 23.5, 23.6): shaker off during recordings, one file per block, lamp kept powered and covered for the dark blocks.

We used Nostoc (Anabaena) sp. PCC 7120 culture in a T25 flask (no biological replicate), medium-only blank in the same session, mixer off. The lamp was different from the one used in the Elodea trials and was not on the sensor's support. In the blank, dark meant lamp switched off. In the culture blocks, dark meant lamp covered but still powered, so light and dark blocks had the same electrical conditions. All blocks lasted ≈ 15 min (90 sweeps).

The block order was reconstructed from the timestamps. The file sample1_light2.csv contains two recordings appended by the logger (12:58–13:13 and 14:11–14:26) and they were separated by timestamp. Note that sample1_light1.csv contains a single sweep (aborted) and was not used.

Every block recorded with the lamp powered exceeded the 2-ADC passive-noise criterion (≈ 20 ADC). By our pre-registered rules the session is not interpretable for biology; the observations below are descriptive.

What the session shows:

- The sensor and the coupling were in good condition. With the lamp off, passive noise was 1.3 ADC and the S2S floor ≈ 4 Hz, the same level as in the Elodea trials. The coupling was stable in every block (|r(peak, time)| ≤ 0.28. No r(peak, centroid) below −0.85 after the first sweeps).
- The mixer problem is solved. With the mixer off, passive noise with the lamp off dropped from 4–11 ADC (23/09) and 5–12 ADC (05/10, shaker on) to 1.3 ADC.
- The noise is electrical, not mechanical. Switching the lamp on produced an instantaneous step and a uniform pedestal across the whole spectrum (≈ +30 ADC on the RMS and above 3200 Hz), also with the transmitter off, and the passive noise rose to ≈ 20 ADC. During the session this was attributed to the lamp a follow-up test showed that the main path is the laptop charger.
- No light effect above the raised floor because the lamp stayed powered in the covered dark blocks, light and dark blocks are electrically comparable, but the S2S floor rose from ≈ 4 to ≈ 11–15 Hz (robust). S2S was similar in light and dark (robust 12–15 vs 13–17 Hz). This is a non-detection at reduced sensitivity and not a clean negative.
- The resonance moved from 1910 Hz (blank) to 1950 Hz (culture), as expected for a different flask and fill (Ch. 21.1).

Follow-up noise test (13/10/2026)

With water only, the same lamp on and dark blocks made with an opaque box, the passive noise was 10.6–11.0 ADC with the laptop on its charger and 1.1–2.0 ADC with the laptop on battery. The main source is therefore a ground loop through the charger (mains -> charger -> USB -> Arduino). With the laptop on battery the lamp itself adds little, and the S2S floor returns to 3–6 Hz. On battery the true mean peak was 110–118 ADC, against 142–144 ADC on the charger: the noise had inflated it.

Corrective actions:

- Laptop on battery for the whole session (or a galvanic USB isolator).
- Lamp outside the box and always on, dark blocks made with the box, shaker off.

- Wait 1–2 min after every light/dark change.
- Re-couple until the peak is ≥ 120–130 ADC at rest, and take the GO / NO-GO decision on battery.
- One file per block.

24. Operating regime

The instrument is validated only inside the following regime. Outside it, a block can pass the gate and still be uninformative (Ch. 14.4, 14.6).

• Vessel: Sarstedt T25 polystyrene flask, fill level marked and constant (Ch. 5, E-11).

• Coupling: peak amplitude at plateau, block mean >= 110 ADC, drop <= 15 % within the block. Vaseline only for sessions shorter than ≈ 20 min after warm-up (Ch. 10, 14.6).

• Bubbling: sparse, intermittent bubbles crossing the region between the discs, above ≈ 4–11 bubbles/min (exploratory limit of detection, Trial 5). A continuous fine stream is a blind regime of the S2S (Ch. 14.4).

• Environment: laptop on battery, passive noise <= 2 ADC [S] in every state, no steps at lamp or mixer switching, mixer off during recordings, temperature <= 25 °C (Ch. 18.5).

• Phase: only the gas phase is seen, i.e. a medium above 100 % O_2 saturation (Ch. 3).

• Comparisons: only inside one session, without remounting the transducers (Ch. 21.3).

PART V: QUALITY, COMPLIANCE AND DEVELOPMENT

25. Error register

This is a list of errors made during development, along with lessons learned. This chapter lists the errors cited in this manual and those encountered subsequently.

25.1 Hardware

ID	Error	Symptom	Cause	Prevention
E-01	Missing bias	Amplitude fixed at ≈ 1023 (or 0)	LM358 pin 3 grounded: the op-amp saturates	10k/10k divider at pin 3, measure 2.5 V before any other checks
E-02	Parasitic splitter	Halved signal	two 10 kΩ to A0	only one protection resistor, count the resistors towards A0
E-03	No couplant	Peak ≈ 10 ADC, flat spectrum	reflection at the air microlayer	always coupling; check peak >= 110 ADC
E-04	Disconnected transducer wires	Intermittent signal	brittle solder joints on the ceramic	soldering on the brass ring, mechanically secure the wires; no traction
E-05	No common ground	High noise, random values	grounds not connected	only one ground node
E-06	Long unshielded cables	High background, structureless waterfall	noise pickup	short cables, NOISE mode, move power supplies away
E-42	Laptop on its charger	Passive noise 10-20 ADC	Ground loop between mains, charger, USB, Arduino	Laptop on battery

25.2 Mechanics and coupling

ID	Error	Symptom	Cause	Prevention
E-07	Vaseline creep	Monotonically decreasing peak (−35…−42%/30 min)	Cold flow under clamp pressure	Silicone grease or gel, GO/NO-GO gate, interruption < 110 ADC
E-08	Overtightening	Accelerated creep, cracked wall	Expulsion of the coupling material; PS stiffness	Screws fully closed + 1/4 turn
E-09	Recording during warm-up	Drifting peak and centroid	Settlement	Wait for plateau
E-10	Transducers touched during session	Jumping peak and centroid	Coupling variation	Intra-session adjustment, restart
E-11	Fluid level fluctuates	Translated resonances	Geometry	Fixed and noted volume, transfer with transducers stopped

25.3 Firmware

ID	Error	Symptom	Cause	Prevention
E-12	Average instead of peak-to-peak	Flat plots with plausible data	Average of a constant centered wave	vmax − vmin; dry/coupled test
E-13	Sweep up to 20 kHz	Unresolved main resonance	Wasted steps in noise	F_MAX = 3500
E-14	Insufficient settlement	Noisy amplitudes	Failure to stabilize	40 ms
E-15	Malformed lines	Lines discarded by the logger	Non-atomic printing or incorrect baud	115 200 on both sides, four fields per line

25.4 Software and acquisition

ID	Error	Symptom	Cause	Prevention
E-16	Serial monitor and logger open together	Port busy	Port exclusivity	Close one before the other
E-17	Failure to wait for reset	First lines lost	Door opening = reset	sleep(2) in the logger, keep the buffered reset header (no input-buffer flush)
E-18	Transitions not logged	Undefinable blocks	No notes	note.md with pc_timestamp
E-19	Clocks not synchronized	Video and audio out of sync	Offset	frame date +%s; merge_asof
E-20	Sessions queued	Last block analysis incorrect	Append	one file per session

25.5 Metrics and analytics

ID	Error	Symptom	Cause	Prevention
E-21	σ of the centroid as an index	σ 28 Hz with near-blank video	slow drift; corr(peak, centroid) −0.97	S2S, QC gate
E-22	Interpretation of a null with a falling peak	"Does not detect" with 20–30 bubbles/min on video	peak −35%	Gate: drop > 15% or corr < −0.85 -> uninterpretable
E-23	Inter-session comparison	−201 Hz "effect"	reassembly drift	Intra-session only
E-24	Expectation of S2S increasing with gas	Abundant gas, S2S at floor level	steady-state blindness	Visual status check; reduced light; absolute metrics on transitions
E-25	Expectation of proportionality with the number of bubbles	Correlation per sweep ≈ 0.6	response ∝ volume/size	Block-level comparisons
E-26	SNR read as bubbles	S2S increasing with decreasing peak	quantization	QC gate
E-27	Autoscaling of graphs	Flat baseline that appears agitated	noise magnification	Fixed axes
E-28	Traffic light indicator	Labels without base	thresholds never calibrated	Disabled, recalibration after validation

ID	Error	Symptom	Cause	Prevention
E-29	Hard-coded thresholds	Indefensible values in the code	placeholder	Three thresholds to be recalibrated, do not cite these as results

25.6 Biology and video reference

ID	Error	Symptom	Cause	Prevention
E-31	Excessive light	Continuous fine plume	High production	100–250 µmol; visual inspection
E-32	Insufficient light or exhausted plant	Nothing at floor	Below LOD	stem re-cut, bicarbonate, pre-adaptation; confirmation video
E-33	Source out of beam	Video full, S2S empty	Geometry	cut at TX–RX level, fixed
E-34	Incomplete darkness	"Dark" with high S2S	Stray light	Darkening, screens off
E-35	No video in darkness	Reversibility not verifiable	Photosynthetic LED = room light	IR illuminator
E-36	False OpenCV detections	≈ 130 bubbles/min	Plant movement/ROI	visual inspection; MAX_AREA; plant fixed
E-37	Temperature > 25°C	Plant suffering, altered solubility	LED heating	Thermometer, distance, ventilation
E-38	BG-11 white assumed as perfect thermal control	Thermal artifact underestimated	Culture absorption	Declare, reversibility and kinetics
E-39	Single biological session	n = 1	Time	>= 3 biological replicates
E-40	Hypochlorite on transducers	Progressive corrosion, intermittent signal	Chlorinated oxidizer on brass/tin/silver	hypochlorite only inside the flask, alcohol on the transducers (Ch. 16)
E-41	Ethanol >= 90% on T25 wall	Opacification, crazing under load	PS sensitivity	70% ethanol, do not rub
E-43	Mixer running during recordings	High variability in light and dark alike; passive noise 4–11 ADC	Mechanical agitation	Mixer off during recordings, mix between blocks

26. Diagnostics

Symptom	**Probable cause**	**Check**	**Action**
No # READY in the serial monitor	Wrong port or baud rate, port busy	Port list; 115 200 baud; other programs	Select the port; close other programs (E-16)
Amplitude ≈ 10 ADC at every step	No coupling, or TX not driven	Couplant; pin 9 and 100 Ω; test_TX_TR	Re-couple (E-03); check the TX branch

Symptom	Probable cause	Check	Action
Flat-topped spectrum at a constant high value	Front-end saturation	Voltages on pins 1 and 3 (Ch. 8.4)	Restore bias (E-01) and feedback
Signal about half of the usual value	Parasitic divider on A0	Resistance pin 1 → A0	Single 10 kΩ (E-02)
Constant, plausible-looking data	Firmware averaging instead of peak-to-peak	measure_amplitude()	Use vmax – vmin (E-12)
Peak keeps falling, centroid rises as its mirror image	Couplant creep or clamp relaxing	Peak trend; r(peak, centroid)	Stop, re-couple, warm up again (E-07)
Step in all features within 1–2 sweeps, synchronous with the light	Lamp interference or laptop-charger ground loop	Same step in the blank; bins above 3200 Hz	Laptop on battery, interference check (Ch. 18.5)
High variability in light and dark alike; large passive noise	Mechanical agitation	Mixer state; P rows	Mixer off during recordings (E-43)
S2S at the floor although bubbles are visible	Continuous fine bubbling, or bubbles outside the beam	Video; bubble path against the beam	Reduce the light; realign (Ch. 14.4)
Dark-block S2S above light-block S2S	S2S computed from centroid_hz, or low SNR	Centroid definition; mean peak	Raw-amplitude centroid, apply the gate
Resonance at a different frequency from the last session	Fill level or mount changed	Fill mark	Expected: compare only inside a session
The board keeps sweeping after the logger is closed	Logger closed abruptly, before it could send X		Unplug the USB cable, or open and close the port: the board restarts idle

27. Some anticipated questions

Dissolved-oxygen probes already exist. Why build this? The probe is invasive and measures the dissolved phase. When bubbling vents the excess O_2, the dissolved concentration reaches a plateau and its slope no longer reflects production (Ch. 3.2). The sensor is external, non-invasive and low-cost and detects the gas phase directly. It complements the probe, it does not replace it.

What does it actually measure? Fluctuations of the gas fraction in the acoustic beam. The attribution to photosynthetic O_2 comes from the experimental design (light dependence, reversibility, controls), not from acoustics (Ch. 2.3).

The centroid is a resonance of your set-up, not a bubble size. Correct. Its absolute position is set by the mount, the information lies in its sweep-to-sweep perturbation (Ch. 4.4, 14.2).

A drift of 281 Hz between mounts invalidates the measurement. It invalidates comparisons between mounts, which the protocol never makes. Within a session the S2S floor is 2.4–5.8 Hz (Ch. 21.3).

Your biological evidence is one block of one session. Yes. Trial 5 is presented as exploratory. Trials 6–8 identified three failure modes (coupling decay, insufficient production, continuous bubbling) and the lamp confound (Ch. 23). The confirmatory session, with pre-registered gate, criteria and controls, is defined in Ch. 18.2.

Did you choose the metric after seeing the data? Yes: the S2S and the gate were introduced after Trial 5, to remove a coupling artefact found there (Ch. 14.3). For this reason Trial 5 is exploratory and the thresholds are fixed for every later session.

Could the light effect be electrical? In Trials 5–8 the lamp raised the passive noise by 1.4–2.4 times. No light effect is claimed from blocks that fail the interference criterion, and the confirmatory protocol includes a blank and a heat-killed control under the same light schedule (Ch. 18.2, 23.4).

28. iGEM compliance

Licences:

- Software (firmware and Python) MIT
- Hardware (circuit, BOM, 3D-printed holder) CERN-OHL-S-2.0
- Documentation and data CC BY 4.0.

Code, firmware, holder and data: https://github.com/Samub432/acoustic-sensor

Wiki: https://2026.igem.wiki/uni-padua-it/.

Organisms used in the validation: Elodea canadensis (non-native in Europe, never released) and Nostoc (Anabaena) sp. PCC 7120; handling and disposal in Ch. 15.3.

Electrical safety: SELV, USB-powered (Ch. 15.1).

Optical safety: Ch. 15.4.

APPENDICES

Appendix A: Glossary

- **ADC:** Analog-to-digital converter. 0–1023 for 0–5 V.
- **Baseline:** Control measurement, in the same session, without the phenomenon under investigation.
- **Blank:** Medium-only control (no organism) recorded with the same light schedule.
- **Block:** Time interval under constant conditions.
- **Coupling agent:** Substance that fills the air microlayer between the transducer and the wall.
- **Creep (cold-flow):** Slow flow of a material under constant load.
- **DO:** Dissolved oxygen.
- **Gate:** Filter criterion: condition that data must satisfy to be valid. Anything that does not satisfy it is discarded.
- **Gate, GO/NO-GO:** Operational gate, at acquisition: determines whether to start or stop recording based on the coupling status (Ch. 17.3).
- **Gate, QC (Quality Control):** Analytical gate, during analysis. It excludes blocks with unreliable coupling before interpretation (Ch. 14.6).
- **Intra-session:** Without moving the transducers. The only context in which absolute values are comparable.
- **LOD:** Limit of detection (≈ 4–11 bubbles/min).
- **Passive noise:** Standard deviation of the P_rown (TX off). It is an indicator of environmental interference.
- **Minnaert:** Relationship between the resonant frequency of a bubble and its radius.
- **Peak (peak ADC):** Maximum amplitude of the raw ADC spectrum; an indicator of the coupling state. peak_ampl_db is the same quantity in dB ($20 \cdot \log_{10}$ of the raw peak, Ch. 14.1).
- **Plateau:** Phase at which the peak ceases to vary after warm-up.
- **RMS:** Root mean square; measure of signal energy.
- **S2S (sweep-to-sweep):** Variation of the centroid between consecutive sweeps: standard deviation of the differences; primary metric (Ch. 14.2).
- **Session:** Everything recorded without moving the transducers.
- **Supersaturation:** Condition (DO > 100 %) above which excess O_2 forms bubbles. The threshold below which the sensor is blind.
- **Sweep:** Sequence of 60 logarithmically spaced frequencies (≈ 999–3427 Hz) lasting ≈ 5 s, followed by ≈ 5 s of passive listening (cadence ≈ 10.0 s).

- **Through-transmission:** Emitter and receiver on opposite sides: the sound passes through the medium once.
- **Trial:** One experimental session, identified by its data folder
- **Void fraction:** Volumetric fraction of gas in the liquid.
- **Warm-up:** Initial phase in which the coupling settles and the peak rises, before the plateau; no recording occurs during this phase.
- **Waterfall:** Colour frequency x time map.

Appendix B: Constants

Sweep (firmware)

Constant	Value	Meaning
"F_MIN"	1000 Hz	Start frequency
"F_MAX"	3500 Hz	End frequency
"N_STEPS"	60	Steps for sweep (log grid)
Settle for step	40ms	Mechanical stabilization before measurement
Measurement window	>3 periods, 1500-4000 µs	Peak-to-peak window
"DWELL_MS"	80ms	Total time for step
"sweep_interval"	5000ms	Passive listening between sweeps
Sweep cadence	10.0s [M]	Sweep-to-sweep period (≈ 5 s sweep + ≈ 5 s passive)

Other modes / ADC

Constant	Value	Meaning
"BURST_FREQ"	8000Hz	Fixed tone in BURST mode
ADC prescaler	16 (≈ 77 kHz)	ADCSRA = (ADCSRA & 0xF8) | 0x04

Interface / acquisition

Constant	Value	Meaning
Baud rate	115200	Serial speed
Reset wait	2s	After opening the port (Arduino reset)
Run separator	Last reset header	after the last # ACOUSTIC BIOREACTOR header

Analysis (acoustic_common.py, 03_features.py)

Constant	Value	Meaning
ANALYSIS_BAND	1000-3200 Hz	Band of every feature
MIN_AMPL_DB	–30 dB	Bins below are ignored by centroid_hz and high_freq_ratio
HIGH_FREQ_THRESH	2000 Hz	Split of high_freq_ratio

Constant	Value	Meaning
MIN_DYN_ADC	20 ADC	Minimum in-band range of a valid sweep
MIN_SWEEP_POINTS	55	Minimum points of a complete sweep
SLOW_WINDOW_N	10 sweeps	_slow components and centroid_sigma_hz
SETTLE_S	180 s	Excluded at the start of LIGHT and DARK files
QC_CORR_THRESH	0.6	|r(peak_ampl_db, centroid_hz)| for QC WARNING
QC_MIN_SWEEPS	20	Below: QC N/A
Coupling gate	r < −0.85; change < −15 %; mean < 110 ADC	Exclusion criteria (Ch. 14.6)
Passive-noise criterion	> 2 ADC [S]	Exclusion criterion, to be pre-registered (Ch. 14.6)

Appendix C: Formulas

Quantity	Formula	Ch.
Sweep grid	f_i = F_MIN * (F_MAX/F_MIN)^(i/N_STEPS), i = 0 … 59	11.1
Minnaert resonance	f_o * a ≈ 3.26 m*Hz, i.e. f_o ≈ 3260 / a[mm] Hz	4.1
Wood's equation	$1/(\rho c^2) = \alpha/(\rho_g c_g^2) + (1-\alpha)/(\rho_l c_l^2)$, with $\rho = \alpha \rho_g + (1-\alpha) \rho_l$	4.2
Centroid for the S2S	c = Σ f·A_raw / Σ A_raw, 1000 ≤ f ≤ 3200 Hz	14.2
S2S	1.4826 * median(|d_i − median(d)|), d_i = c_{i+1} − c_i	14.2
Robust S2S	1.4826 * median(|d_i − median(d)|)	
centroid_hz	Σ f·A_n / Σ A_n over the bins with $20 \cdot \log_{10}$ A_n > −30 dB; A_n = (A − min A) / max(A − min A)	14.1
rms_abs_db	$10*\log_{10}$(mean A_{raw}^2)	14.1
peak_ampl_db	$20*\log_{10}$(max A_raw); 110 ADC = 40.8 dB, 150 ADC = 43.5 dB	14.1
high_freq_ratio	Σ_{f>2000} A_n / Σ A_n over the valid bins	14.1
centroid_sigma_hz	Standard deviation of centroid_hz over a centred 10-sweep window	14.1
Peak change	(mean of the last 3 peaks − mean of the first 3) / mean of the first 3	14.2
Passive noise	Median over the pauses of a block of std(P rows of the pause)	18.5
Bracketed difference	d = (X_light,before + X_light,after) / 2 − X_dark	19.3